Project No.	K1AU3

LMYP-1 PROJECT

Mass Balance Diagram

For Approval

FORMOSA HEAVY INDUSTRIES CORP.
CO-GENERATION DEPARTMENT

Total 2+24 sheets, including this cover

Rev. no.	Revision description	Revised	Check	Approved

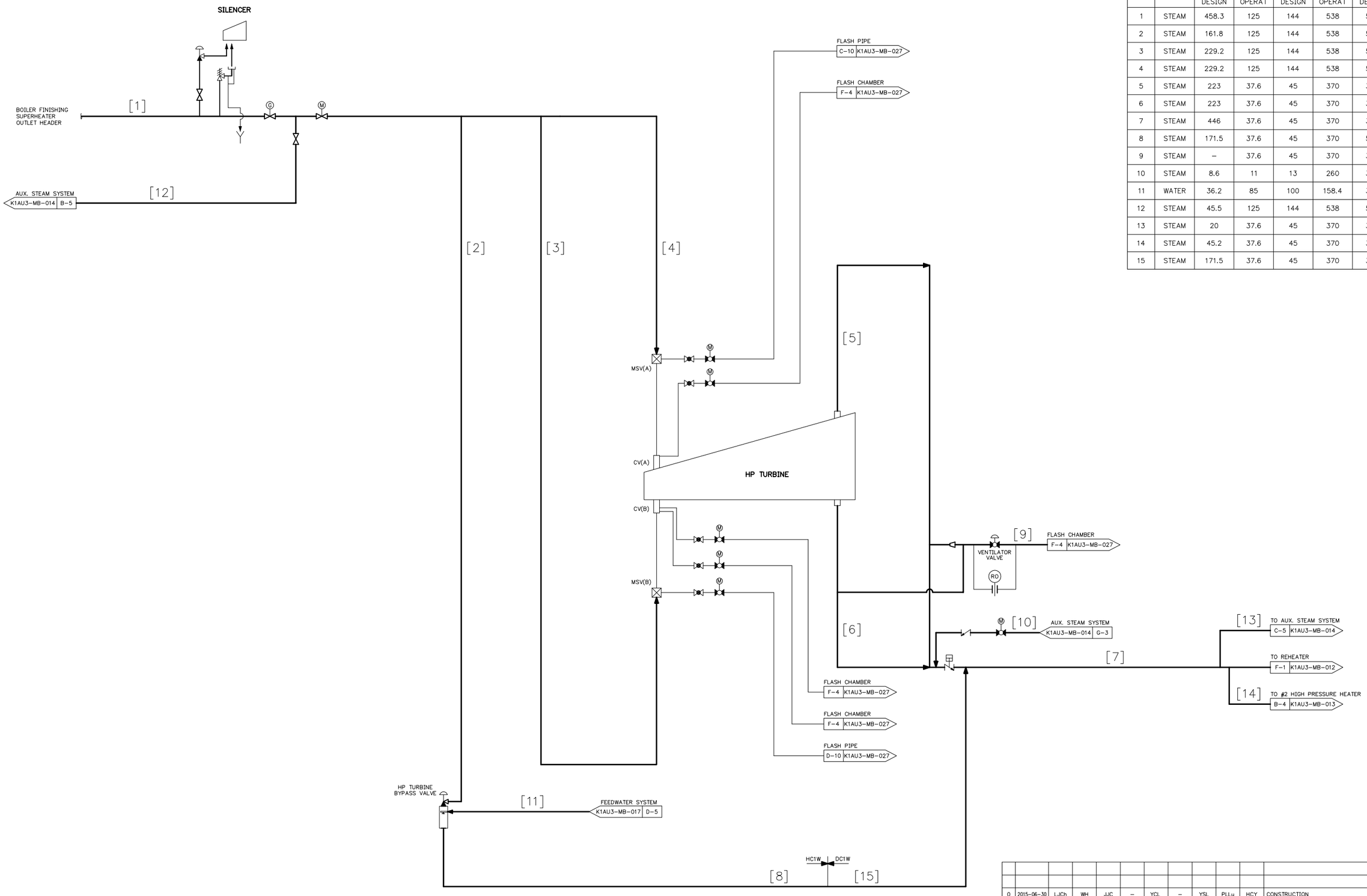
BY	Rex Li	2015.07.13	FORMOSA HEAVY INDUSTRIES CORP. ISSUED JUL 13 2015 COGEN DEPT · BASIC DESIGN	Document No.
CHK.	Rex Li	2015.07.13		K1AU3-A2-0-001
APPR.	Kimble Lin	2015.07.13		Rev. A

LMYP-1 PROJECT

DRAWING LIST---MASS BALANCE DIAGRAM

Item	Drawing No.	Title	REVISION & ISSUE DATE to SMC			
			2015/7/13			
1	K1AU4-MB-011	MASS BALANCE DIAGRAM - MAIN STEAM SYSTEM	0			
2	K1AU4-MB-012	MASS BALANCE DIAGRAM - COLD REHEAT AND HOT REHEAT STEAM SYSTEM	0			
3	K1AU4-MB-013	MASS BALANCE DIAGRAM - EXTRACTION STEAM SYSTEM	0			
4	K1AU4-MB-014	MASS BALANCE DIAGRAM - AUXILIARY STEAM, PACKING STEAM AND CONDENSER AIR REMOVAL SYSTEM	0			
5	K1AU4-MB-015	MASS BALANCE DIAGRAM - STEAM SOOT BLOWING SYSTEM	0			
6	K1AU4-MB-016	MASS BALANCE DIAGRAM - CONDENSATE SYSTEM	0			
7	K1AU4-MB-017	MASS BALANCE DIAGRAM - FEEDWATER SYSTEM	0			
8	K1AU4-MB-018	MASS BALANCE DIAGRAM - HEATER VENT AND DRAIN SYSTEM	0			
9	K1AU4-MB-019	MASS BALANCE DIAGRAM - BOILER VENT AND DRAIN SYSTEM	0			
10	K1AU4-MB-020	MASS BALANCE DIAGRAM - SEAWATER SYSTEM	0			
11	K1AU4-MB-021	MASS BALANCE DIAGRAM - AUXILIARY COOLING WATER SYSTEM	0			
12	K1AU4-MB-022	MASS BALANCE DIAGRAM - DIESEL OIL SYSTEM	0			
13	K1AU4-MB-023	MASS BALANCE DIAGRAM - COMPRESSED AIR SYSTEM (SHEET NO. 1/2)	0			
14	K1AU4-MB-023	MASS BALANCE DIAGRAM - COMPRESSED AIR SYSTEM (SHEET NO. 2/2)	0			
15	K1AU4-MB-024	MASS BALANCE DIAGRAM - MAKE-UP WATER SYSTEM	0			
16	K1AU4-MB-025	MASS BALANCE DIAGRAM - INDUSTRY WATER SYSTEM	0			
17	K1AU4-MB-026	MASS BALANCE DIAGRAM - POTABLE WATER SYSTEM	0			
18	K1AU4-MB-027	MASS BALANCE DIAGRAM - MISCELLANEOUS VENTS AND DRAINS SYSTEM	0			
19	K1AU4-MB-051	MASS BALANCE DIAGRAM - COMBUSTION AIR / FLUE GAS SYSTEM	A			
20	K1AU4-MB-052	MASS BALANCE DIAGRAM - PRIMARY AIR SYSTEM	A			
21	K1AU4-MB-053	MASS BALANCE DIAGRAM - SECONDARY AIR SYSTEM	A			
22	K1AU4-MB-054	MASS BALANCE DIAGRAM - FLUIDIZING AIR SYSTEM (SHEET NO. 1/3)	A			
23	K1AU4-MB-054	MASS BALANCE DIAGRAM - FLUIDIZING AIR SYSTEM (SHEET NO. 2/3)	A			
24	K1AU4-MB-054	MASS BALANCE DIAGRAM - FLUIDIZING AIR SYSTEM (SHEET NO. 3/3)	A			

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	STEAM	458.3	125	144	538	550
2	STEAM	161.8	125	144	538	550
3	STEAM	229.2	125	144	538	550
4	STEAM	229.2	125	144	538	550
5	STEAM	223	37.6	45	370	395
6	STEAM	223	37.6	45	370	395
7	STEAM	446	37.6	45	370	395
8	STEAM	171.5	37.6	45	370	550
9	STEAM	–	37.6	45	370	395
10	STEAM	8.6	11	13	260	300
11	WATER	36.2	85	100	158.4	395
12	STEAM	45.5	125	144	538	550
13	STEAM	20	37.6	45	370	395
14	STEAM	45.2	37.6	45	370	395
15	STEAM	171.5	37.6	45	370	395

REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
0	2015-06-30	LJCh	WH	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	WH	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

APPROVALS

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SHEET CONTENT

MASS BALANCE DIAGRAM
MAIN STEAM SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

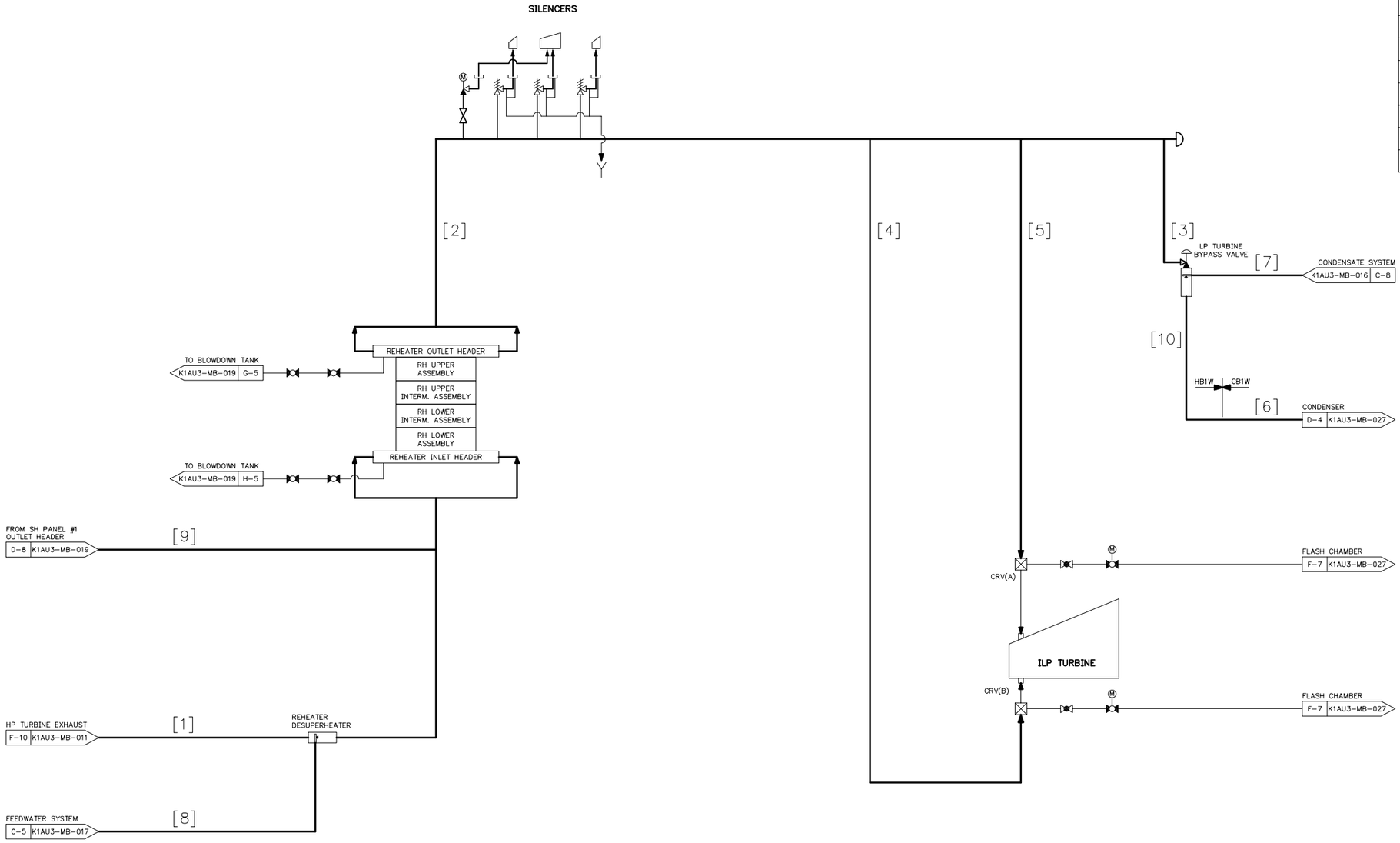
吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO. K1AU3-MB-011 | REV. 0 | SCALE NONE | SHT.NO. 1/1 | JOB NO. 1881

Drawing name: D:\draw\1881\ k1au3-mb-011-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	STEAM	401	37.6	45	370	395
2	STEAM	401	33.8	45	538	550
3	STEAM	171.8	33.8	45	538	550
4	STEAM	200.5	33.8	45	538	550
5	STEAM	200.5	33.8	45	538	550
6	STEAM	217	10.7	15	187	280
7	WATER	45.2	18	24	40	65.6
8	WATER	21	37.6	45	158.4	395
9	STEAM	25	37.6	45	360	395
10	STEAM	217	10.7	15	541	550



REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
					APPROVALS						
0	2015-06-30	LJCh	WH	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	WH	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

COPY

ENGINEER

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SHEET CONTENT

MASS BALANCE
COLD REHEAT AND
HOT REHEAT STEAM SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

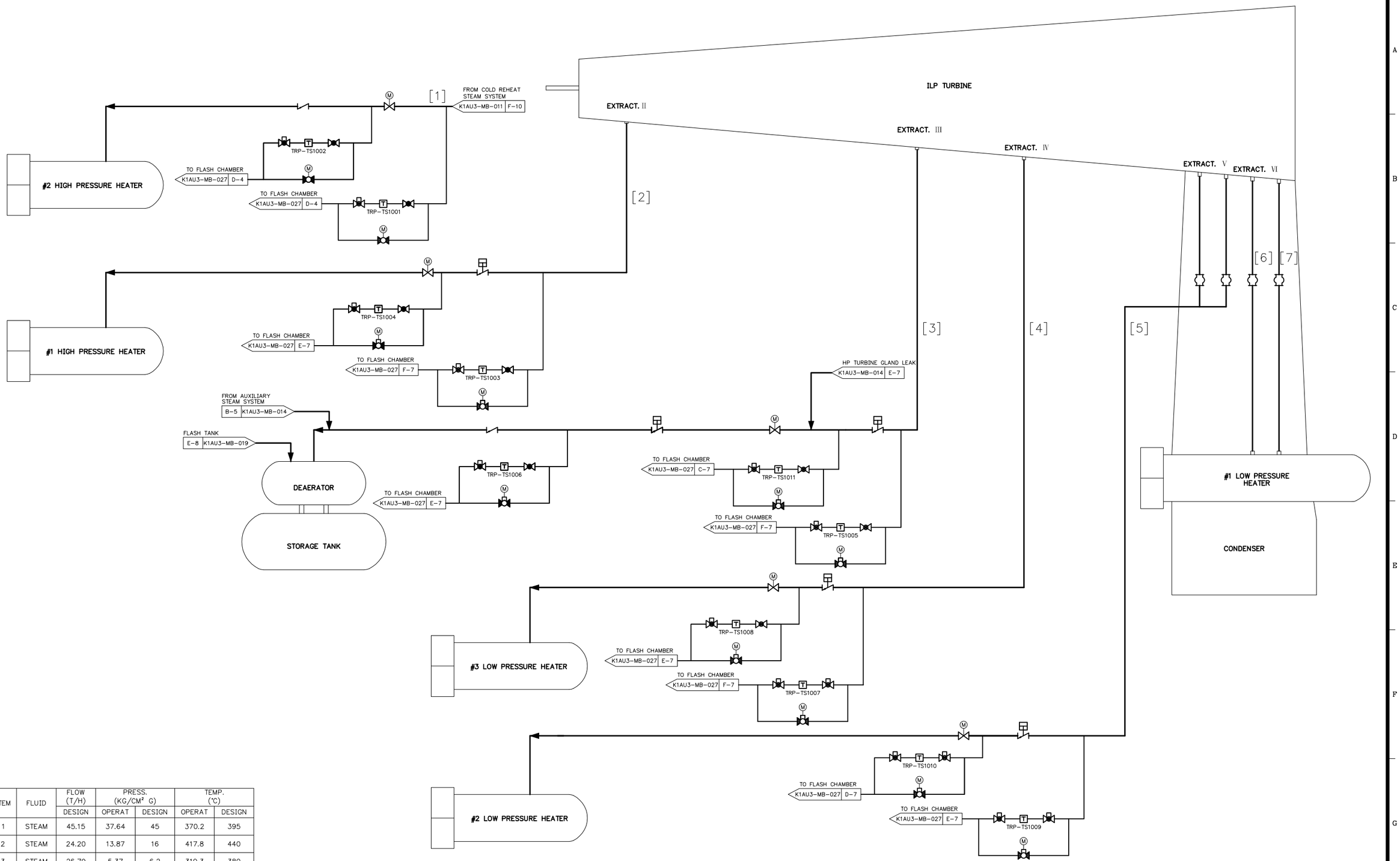
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-012	REV.	SCALE	NONE	SHT.NO.
		0	JOB NO.	1881	1/1

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (℃)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	STEAM	45.15	37.64	45	370.2	395
2	STEAM	24.20	13.87	16	417.8	440
3	STEAM	26.79	5.37	6.2	310.3	380
4	STEAM	15.25	0.91	3.6	181.1	285
5	STEAM	12.27	−0.17	3.6	105.9	285
6	STEAM	8.42	−0.63	3.6	73.7	285
7	STEAM	8.42	−0.63	3.6	73.7	285

REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
					APPROVALS						
0	2015-06-30	LJCh	HGL	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	MJ	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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SHEET CONTENT

MASS BALANCE DIAGRAM
EXTRACTION STEAM SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

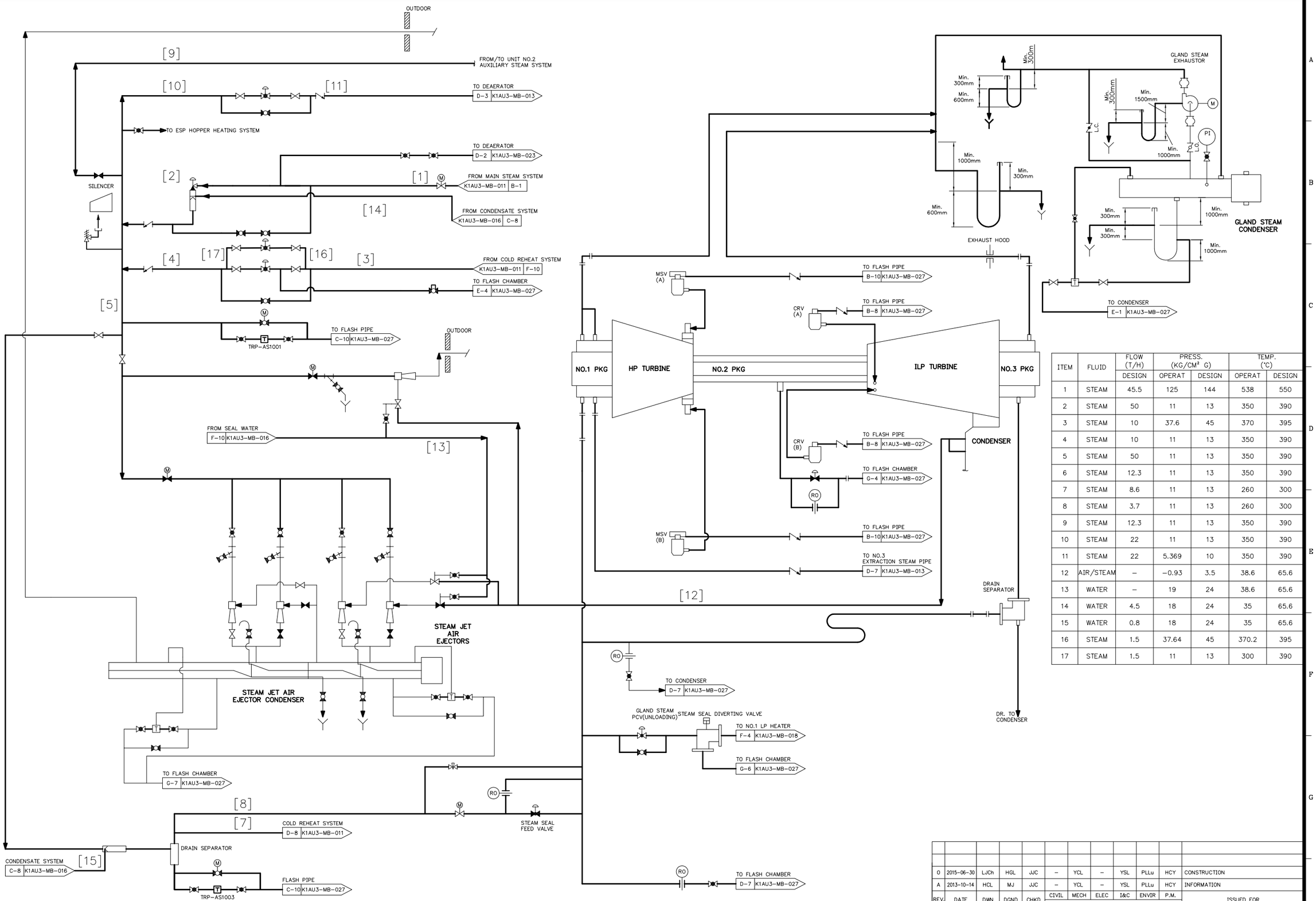
SMC Consolidated Power Corporation
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1550 METRO MANILA, PHILIPPINES

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GIBSIN Engineers, Ltd.
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DWG NO. K1AU3-MB-013 | REV. 0 | SCALE NONE | JOB NO. 1881 | SHT.NO. 1/1

Drawing name: D:\draw\1881\ k1au3-mb-013-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	STEAM	45.5	125	144	538	550
2	STEAM	50	11	13	350	390
3	STEAM	10	37.6	45	370	395
4	STEAM	10	11	13	350	390
5	STEAM	50	11	13	350	390
6	STEAM	12.3	11	13	350	390
7	STEAM	8.6	11	13	260	300
8	STEAM	3.7	11	13	260	300
9	STEAM	12.3	11	13	350	390
10	STEAM	22	11	13	350	390
11	STEAM	22	5.369	10	350	390
12	AIR/STEAM	–	−0.93	3.5	38.6	65.6
13	WATER	–	19	24	38.6	65.6
14	WATER	4.5	18	24	35	65.6
15	WATER	0.8	18	24	35	65.6
16	STEAM	1.5	37.64	45	370.2	395
17	STEAM	1.5	11	13	300	390

REV.	DATE	DWN	DGND	CHKD	APPROVALS CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
0	2015-06-30	LJCh	HGL	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	MJ	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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SHEET CONTENT

MASS BALANCE DIAGRAM AUXILIARY STEAM,TURBINE PACKING STEAM AND CONDENSER AIR REMOVAL SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT

LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

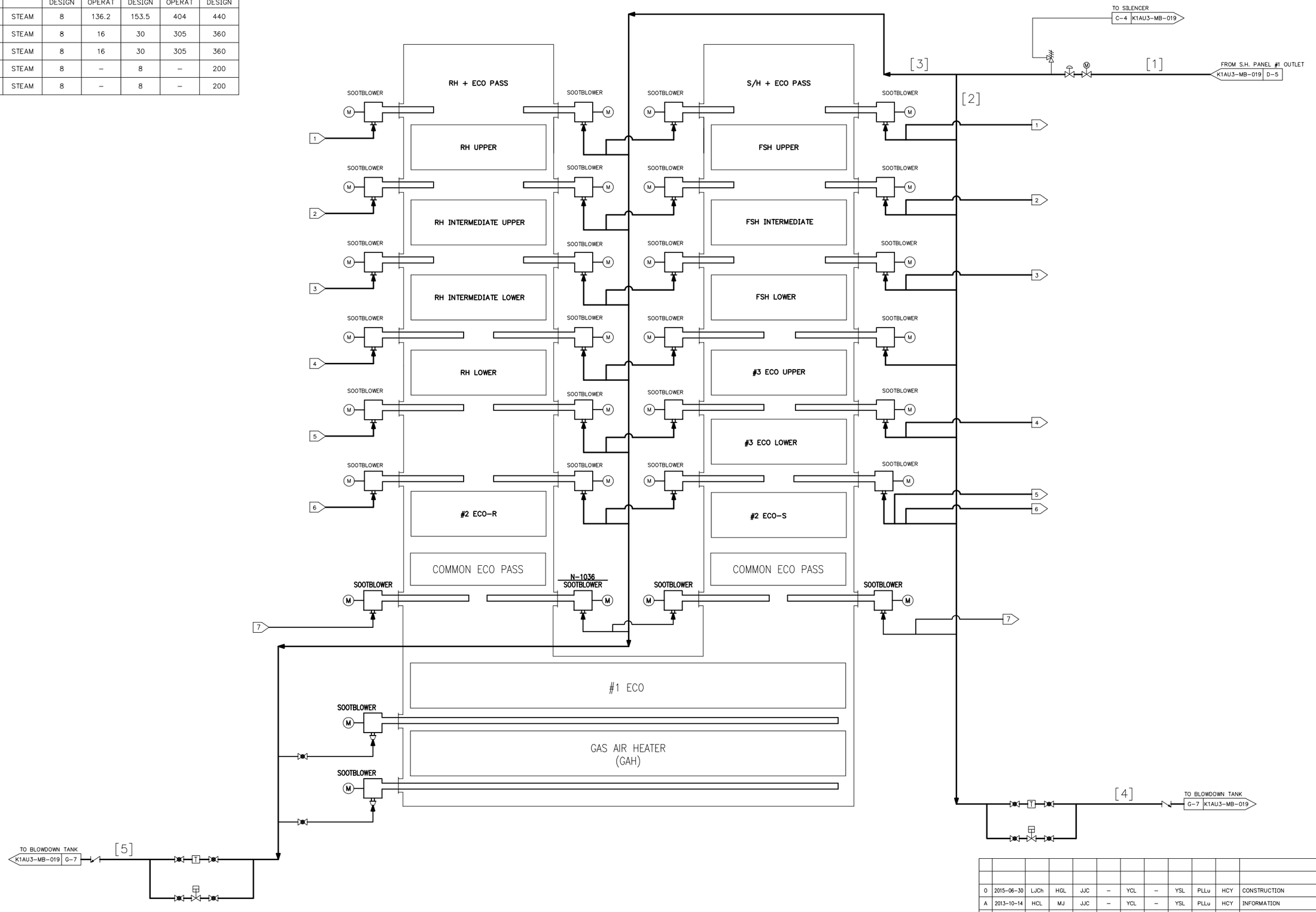
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GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO. K1AU3-MB-014 | REV. 0 | SCALE NONE | JOB NO. 1881 | SHT.NO. 1/1

Drawing name: D:\draw\1881\ k1au3-mb-014-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	STEAM	8	136.2	153.5	404	440
2	STEAM	8	16	30	305	360
3	STEAM	8	16	30	305	360
4	STEAM	8	–	8	–	200
5	STEAM	8	–	8	–	200



REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
					APPROVALS						
0	2015-06-30	LJCh	HGL	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	MJ	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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SHEET CONTENT

MASS BALANCE DIAGRAM

STEAM SOOT BLOWING SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT

LIMAY, BATAAN, PHILIPPINES

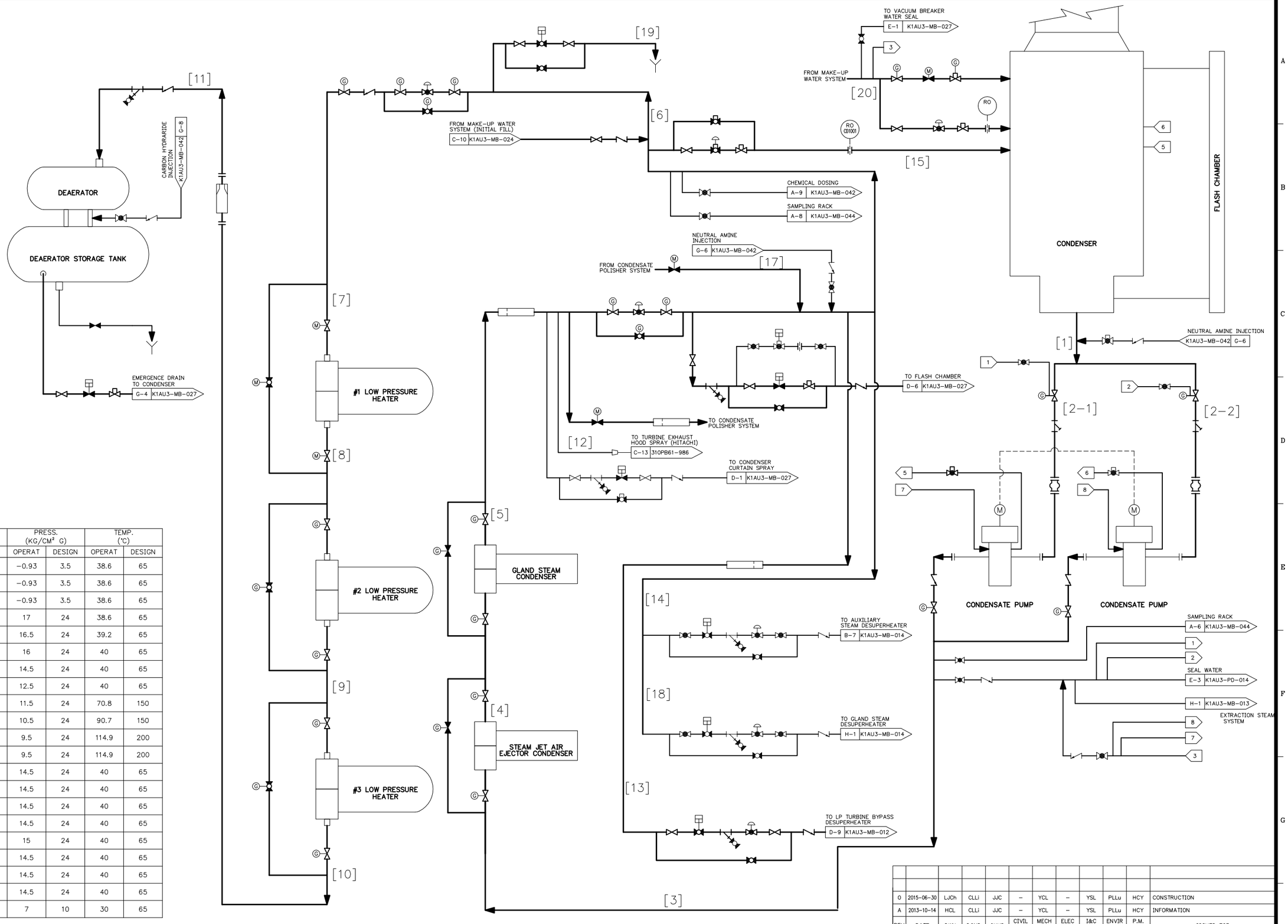
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40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-015	REV.	SCALE	NONE	SHT.NO.
		0	JOB NO.	1881	1/1

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (℃)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	WATER	400	−0.93	3.5	38.6	65
2−1	WATER	400	−0.93	3.5	38.6	65
2−2	WATER	400	−0.93	3.5	38.6	65
3	WATER	400	17	24	38.6	65
4	WATER	400	16.5	24	39.2	65
5	WATER	400	16	24	40	65
6	WATER	400	14.5	24	40	65
7	WATER	400	12.5	24	40	65
8	WATER	400	11.5	24	70.8	150
9	WATER	400	10.5	24	90.7	150
10	WATER	400	9.5	24	114.9	200
11	WATER	400	9.5	24	114.9	200
12	WATER	5.3	14.5	24	40	65
13	WATER	45.2	14.5	24	40	65
14	WATER	12.5	14.5	24	40	65
15	WATER	120	14.5	24	40	65
16	WATER	200	15	24	40	65
17	WATER	200	14.5	24	40	65
18	WATER	8	14.5	24	40	65
19	WATER	80	14.5	24	40	65
20	WATER	70	7	10	30	65

REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	
					APPROVALS						
0	2015-06-30	LJCh	CLLi	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	CLLi	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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T.I.N.	:

SHEET CONTENT

MASS BLANCE DIAGRAM
CONDENSATE SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

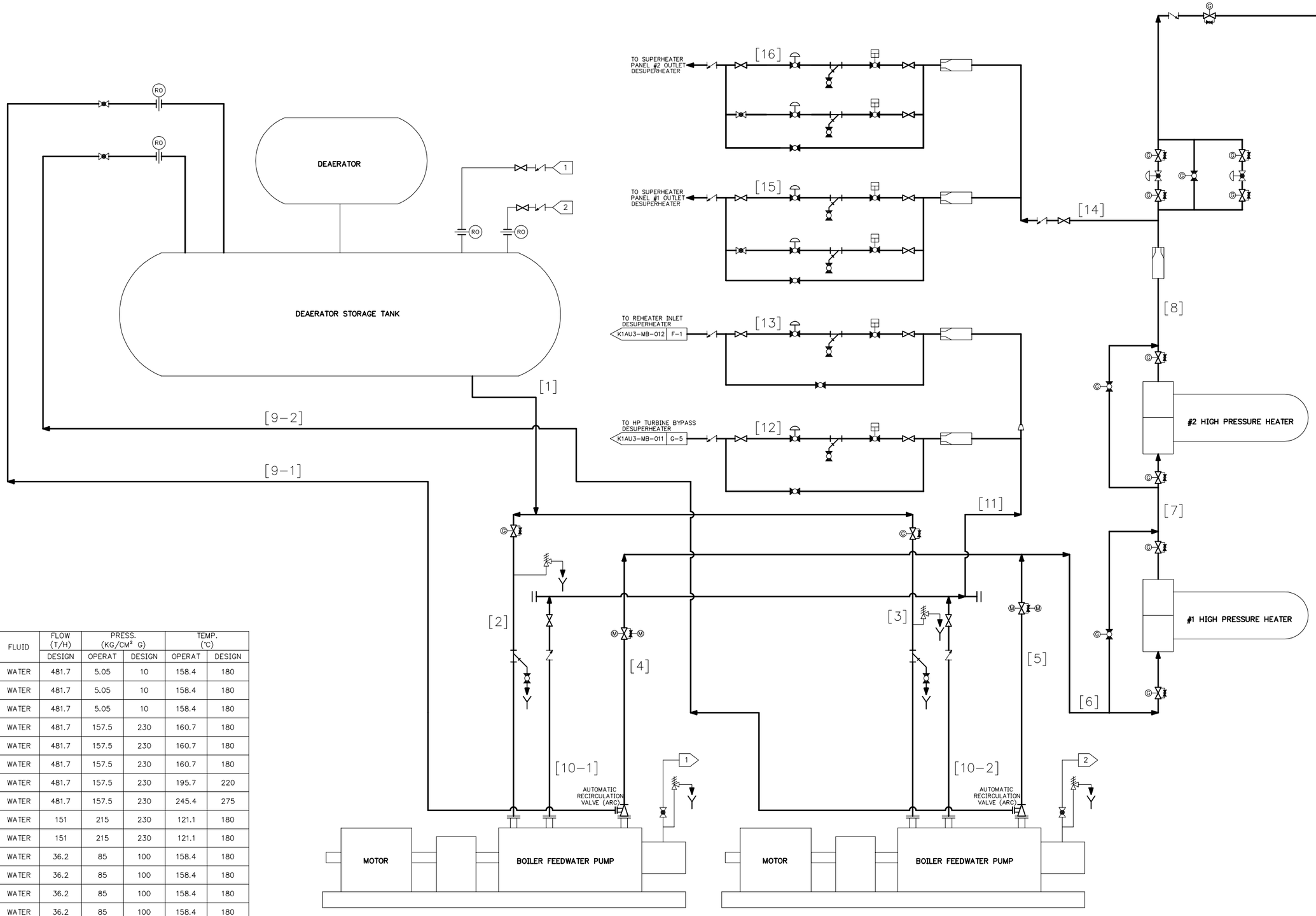
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40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-016	REV.	SCALE	NONE	SHT.NO.
		0	JOB NO.	1881	1/1

Drawing name: D:\draw\1881\ k1au3-mb-016-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (℃)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	WATER	481.7	5.05	10	158.4	180
2	WATER	481.7	5.05	10	158.4	180
3	WATER	481.7	5.05	10	158.4	180
4	WATER	481.7	157.5	230	160.7	180
5	WATER	481.7	157.5	230	160.7	180
6	WATER	481.7	157.5	230	160.7	180
7	WATER	481.7	157.5	230	195.7	220
8	WATER	481.7	157.5	230	245.4	275
9-1	WATER	151	215	230	121.1	180
9-2	WATER	151	215	230	121.1	180
10-1	WATER	36.2	85	100	158.4	180
10-2	WATER	36.2	85	100	158.4	180
11	WATER	36.2	85	100	158.4	180
12	WATER	36.2	85	100	158.4	180
13	WATER	21	85	100	158.4	180
14	WATER	70	157.5	230	245.4	275
15	WATER	35	157.5	230	245.4	275
16	WATER	35	157.5	230	245.4	275

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REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	
					APPROVALS						ISSUED FOR
0	2015-06-30	LJCh	HGL	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	MJ	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

ENGINEER

REG. NO. :
P.T.R. NO. :
DATE :
PLACE :
T.I.N. :

SHEET CONTENT

MASS BALANCE DIAGRAM
FEEDWATER SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

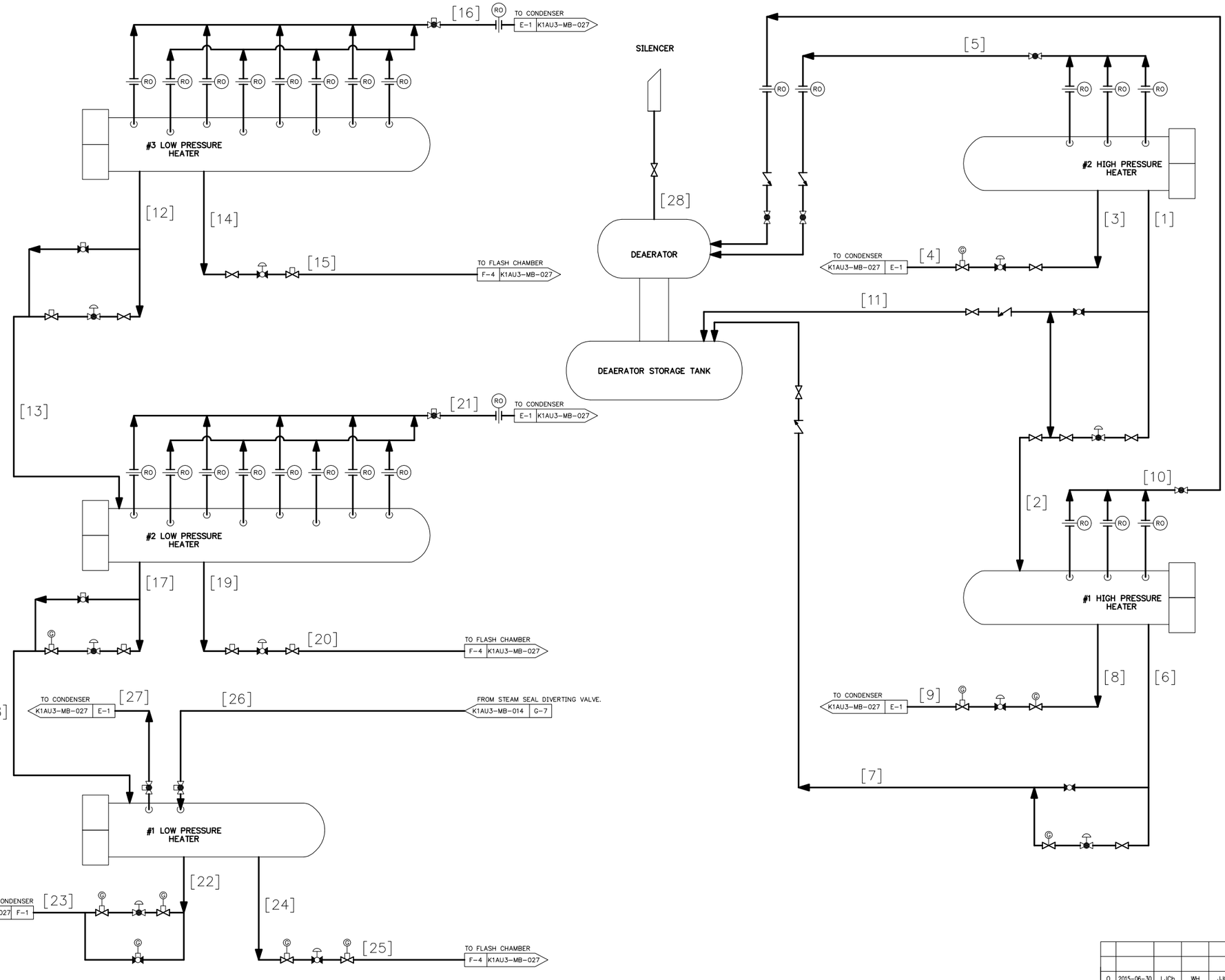
OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-017	REV.	SCALE	NONE	SHT.NO.
		0	JOB NO.	1881	1/1

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	WATER	45.147	36.48	45&F.V.	201	260
2	WATER	45.147	13.12	16&F.V.	201	260
3	WATER	45.147	36.48	45&F.V.	201	260
4	WATER	45.147	−0.93	1.055&F.V.	201	260
5	AIR& VAPOR	–	36.48	45&F.V.	201	260
6	WATER	69.347	13.12	16&F.V.	166	210
7	WATER	69.347	5.05	6.2	166	210
8	WATER	69.347	13.12	16&F.V.	166	210
9	WATER	69.347	−0.93	1.055&F.V.	166	210
10	AIR& VAPOR	–	13.12	16&F.V.	166	210
11	WATER	45.147	5.05	6.2	201	260
12	WATER	15.246	0.82	3.6&F.V.	96	200
13	WATER	15.246	−0.215	3.6&F.V.	96	200
14	WATER	15.246	0.82	3.6&F.V.	96	200
15	WATER	15.246	−0.93	1.055&F.V.	96	200
16	AIR& VAPOR	–	0.82	3.6&F.V.	96	200
17	WATER	27.516	−0.4	3.6&F.V.	76	150
18	WATER	27.516	−0.646	3.6&F.V.	76	150
19	WATER	27.516	−0.25	3.6&F.V.	76	150
20	WATER	27.516	−0.93	1.055&F.V.	76	150
21	AIR& VAPOR	–	−0.215	3.6&F.V.	76	150
22	WATER	45.377	−0.8	3.6&F.V.	45	150
23	WATER	45.377	−0.93	1.055&F.V.	45	150
24	WATER	45.377	−0.7	3.6&F.V.	45	150
25	WATER	45.377	−0.93	1.055&F.V.	45	150
26	STEAM	–	0.5	2&F.V.	350	400
27	WATER	–	−0.646	3.6&F.V.	45	150
28	AIR& VAPOR	–	5.05	6.2&F.V.	310	380

REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
					APPROVALS						
0	2015-06-30	LJCh	WH	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	WH	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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DATE :
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SHEET CONTENT

MASS BALANCE DIAGRAM HEATER VENT AND DRAIN SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT

LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

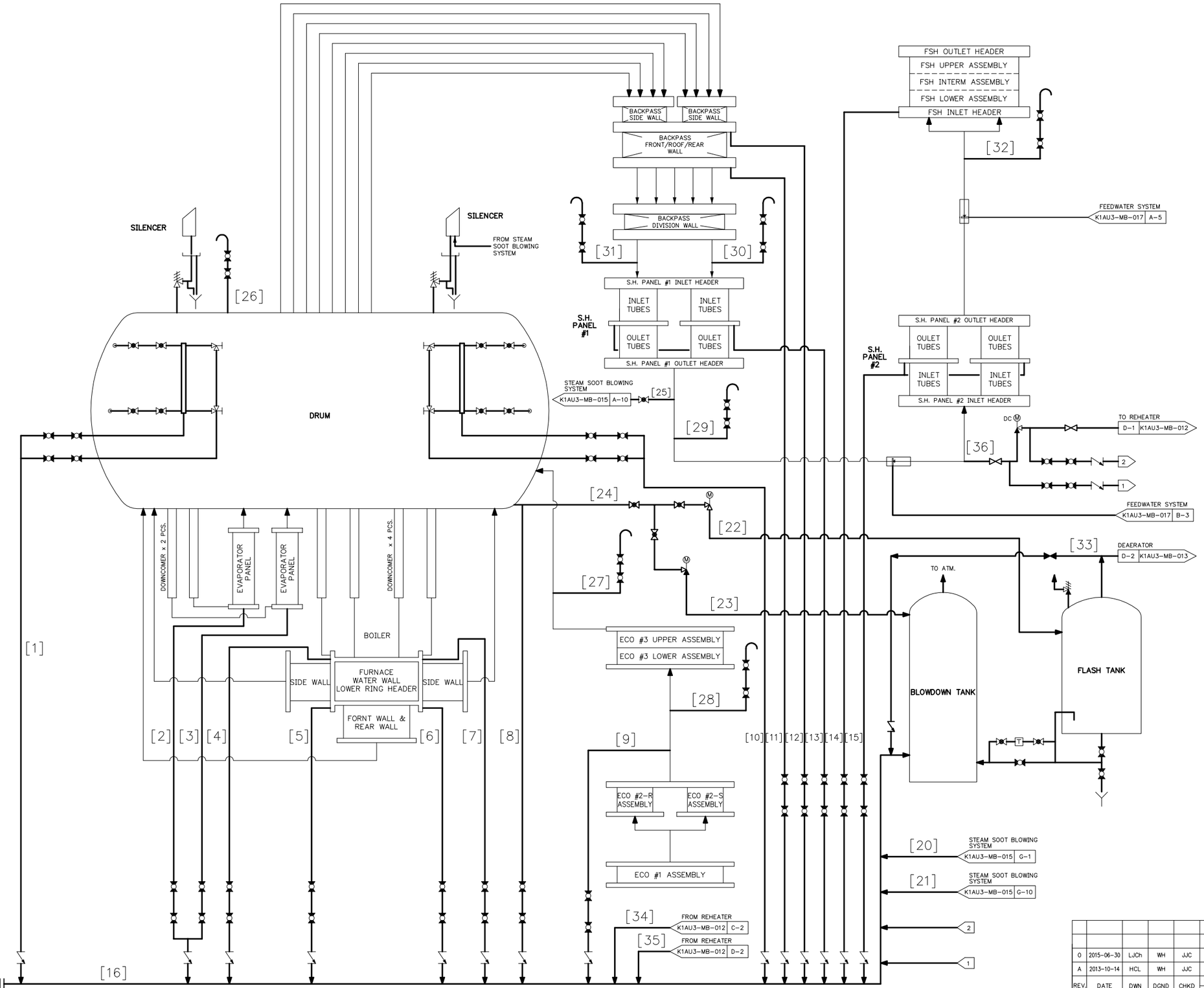
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-018	REV.	SCALE	NONE	SHT.NO.
		0	JOB NO.	1881	1/1

Drawing name: D:\draw\1881\ k1au3-mb-018-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	WATER	–	5.5	8	160	280
2	WATER	–	142.5	155.4	339	343
3	WATER	–	142.5	155.4	339	343
4	WATER	–	142.5	155.4	339	343
5	WATER	–	142.5	155.4	339	343
6	WATER	–	142.5	155.4	339	343
7	WATER	–	142.5	155.4	339	343
8	WATER	–	142.5	155.4	339	343
9	WATER	–	144.5	157	279	343
10	WATER	–	5.5	8	160	280
11	STEAM	–	139.5	153.5	343	377
12	STEAM	–	140.5	153.5	340	377
13	STEAM	–	137	153.5	374	452
14	STEAM	–	130.5	153.5	500	521
15	STEAM	–	134.1	153.5	428	482
16	WATER	–	5.5	8	160	280
20	WATER	–	–	8	–	200
21	WATER	–	–	8	–	200
22	WATER	15	5.5	63	160	280
23	WATER	50	5.5	63	160	280
24	WATER	50	142.5	155.4	339	343
25	STEAM	8	136.2	153.5	404	440
26	AIR& VAPOR	–	142.5	155.4	339	343
27	AIR& VAPOR	–	143.8	157	303	343
28	AIR& VAPOR	–	144.5	157	279	343
29	AIR& VAPOR	–	136.2	153.5	404	440
30	AIR& VAPOR	–	138.5	153.5	345	360
31	AIR& VAPOR	–	138.5	153.5	345	360
32	STEAM	–	130.5	153.5	500	521
33	STEAM	6.5	5	12.5	158	200
34	WATER	–	38	45	370	395
35	WATER	–	38	45	538	550
36	STEAM	25	136.2	153.5	404	440

REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
					APPROVALS						
0	2015-06-30	LJCh	WH	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	WH	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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SHEET CONTENT

MASS BALANCE DIAGRAM
BOILER VENT AND DRAIN SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

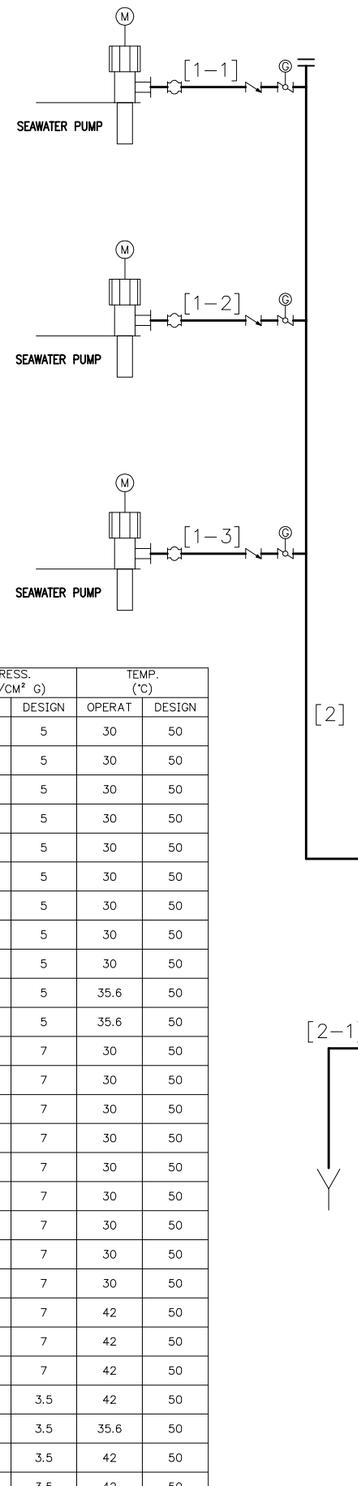
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

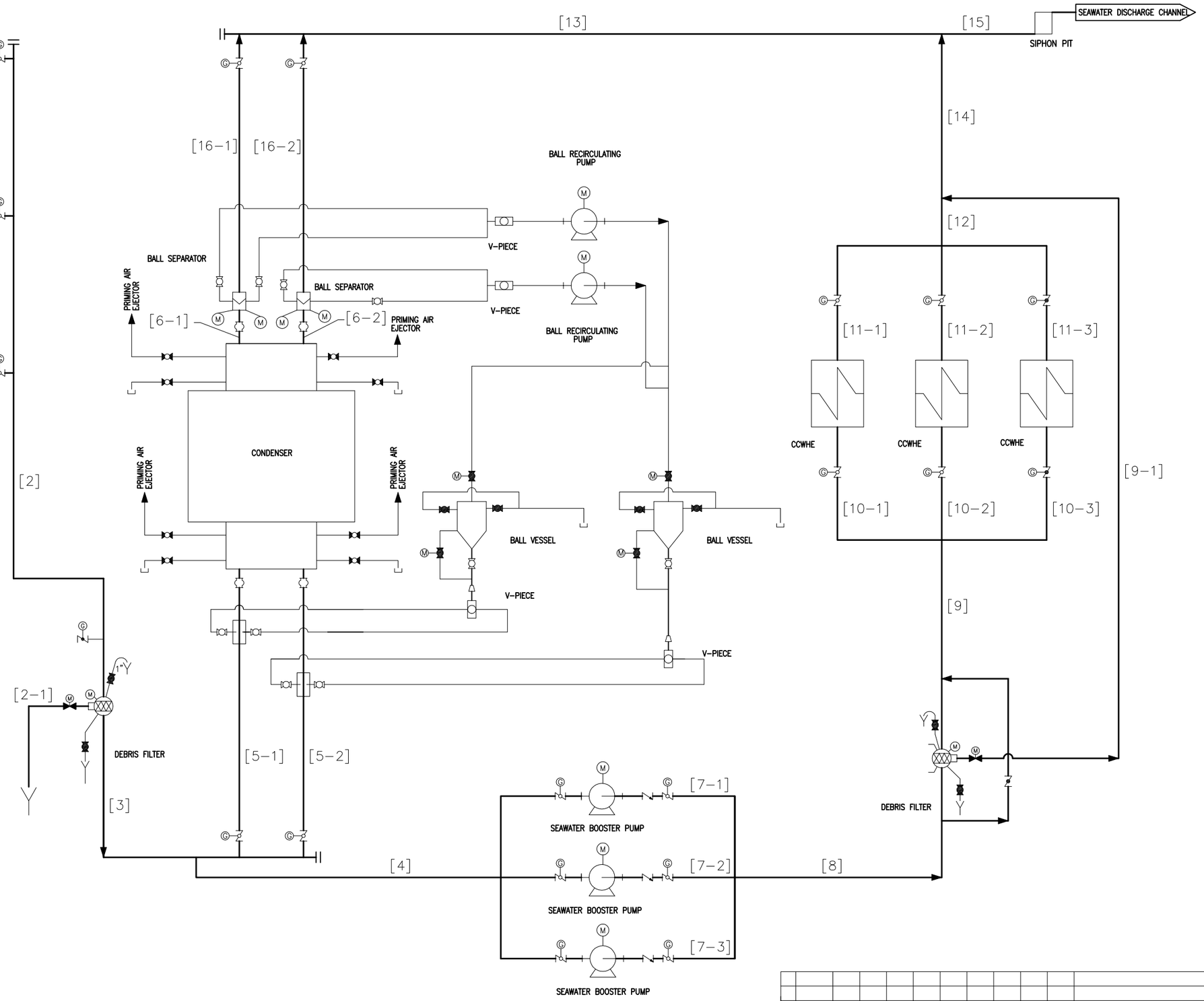
DWG NO. K1AU3-MB-019 | REV. 0 | SCALE NONE | JOB NO. 1881 | SHT.NO. 1/1

Drawing name: D:\draw\1881\ k1au3-mb-019-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (℃)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1-1	SEAWATER	18900	2.8	5	30	50
1-2	SEAWATER	18900	2.8	5	30	50
1-3	SEAWATER	18900	2.8	5	30	50
2	SEAWATER	37800	2.8	5	30	50
2-1	SEAWATER	1890	1	5	30	50
3	SEAWATER	37800	2.5	5	30	50
4	SEAWATER	2000	2.5	5	30	50
5-1	SEAWATER	17900	2.5	5	30	50
5-2	SEAWATER	17900	2.5	5	30	50
6-1	SEAWATER	17900	2	5	35.6	50
6-2	SEAWATER	17900	2	5	35.6	50
7-1	SEAWATER	1000	4	7	30	50
7-2	SEAWATER	1000	4	7	30	50
7-3	SEAWATER	1000	4	7	30	50
8	SEAWATER	2000	4	7	30	50
9	SEAWATER	2000	3.5	7	30	50
9-1	SEAWATER	100	3.5	7	30	50
10-1	SEAWATER	1000	3.5	7	30	50
10-2	SEAWATER	1000	3.5	7	30	50
10-3	SEAWATER	1000	3.5	7	30	50
11-1	SEAWATER	1000	2.5	7	42	50
11-2	SEAWATER	1000	2.5	7	42	50
11-3	SEAWATER	1000	2.5	7	42	50
12	SEAWATER	2000	2.5	3.5	42	50
13	SEAWATER	37800	2.2	3.5	35.6	50
14	SEAWATER	2000	2.2	3.5	42	50
15	SEAWATER	37800	2.2	3.5	42	50
16-1	SEAWATER	17900	2	5	35.6	50
16-2	SEAWATER	17900	2	5	35.6	50



REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
					APPROVALS						
0	2015-06-30	LJCh	CLLi	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	YWHu	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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SHEET CONTENT

MASS BALANCE DIAGRAM
SEAWATER SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

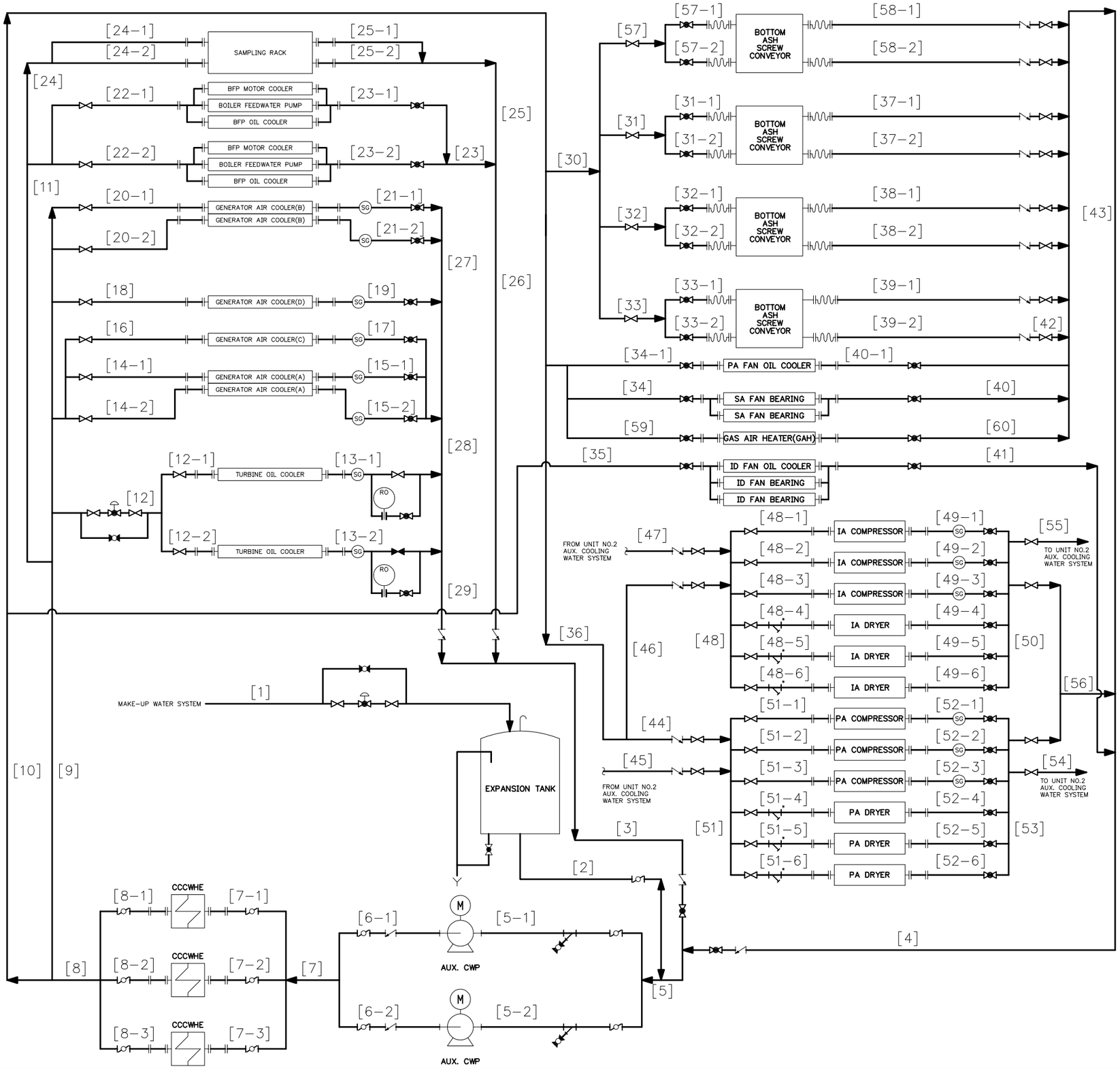
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40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-020	REV.	0	SCALE	NONE	SHT.NO.	1/1
				JOB NO.	1881		

Drawing name: D:\draw\1881\ k1au3-mb-020-0.dwg Dated: 2015-07-07 4:45 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)		ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN			DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	WATER	20	3	10	30	65	40-1	WATER	2	4	10	37	65
2	WATER	0	3	10	30	65	41	WATER	59	4	10	43	65
3	WATER	850	5.5	10	43	65	43	WATER	80	6	10	62	80
4	WATER	415	3	10	55	65	44	WATER	200	6	10	34	65
5	WATER	1500	3	10	50	65	45	WATER	200	6	10	34	65
5-1	WATER	1500	3	10	50	65	46	WATER	70	6	10	34	65
5-2	WATER	1500	3	10	50	65	47	WATER	70	6	10	34	65
6-1	WATER	1500	8	10	50	65	48	WATER	70	6	10	34	65
6-2	WATER	1500	8	10	50	65	48-1	WATER	29	6	10	34	65
7	WATER	1500	7.8	10	50	65	48-2	WATER	29	6	10	34	65
7-1	WATER	750	7.8	10	50	65	48-3	WATER	29	6	10	34	65
7-2	WATER	750	7.8	10	50	65	48-4	WATER	6	6	10	34	65
7-3	WATER	750	7.8	10	50	65	48-5	WATER	6	6	10	34	65
8	WATER	1500	7.2	10	34	65	48-6	WATER	6	6	10	34	65
8-1	WATER	750	7.2	10	34	65	49-1	WATER	29	5	10	43	65
8-2	WATER	750	7.2	10	34	65	49-2	WATER	29	5	10	43	65
8-3	WATER	750	7.2	10	34	65	49-3	WATER	29	5	10	43	65
9	WATER	850	7	10	34	65	49-4	WATER	6	5.75	10	40	65
10	WATER	415	7	10	34	65	49-5	WATER	6	5.75	10	40	65
11	WATER	110	6.5	10	34	65	49-6	WATER	6	5.75	10	40	65
12	WATER	170	5	10	34	65	50	WATER	70	5.7	10	43	65
12-1	WATER	170	5	10	34	65	51	WATER	200	6	10	34	65
12-2	WATER	170	5	10	34	65	51-1	WATER	80	6	10	34	65
13-1	WATER	170	4.75	10	41	65	51-2	WATER	80	6	10	34	65
13-2	WATER	170	4.75	10	41	65	51-3	WATER	80	6	10	34	65
14-1	WATER	76.5	6.5	10	34	65	51-4	WATER	18	6	10	34	65
14-2	WATER	76.5	6.5	10	34	65	51-5	WATER	18	6	10	34	65
15-1	WATER	76.5	6	10	42	65	51-6	WATER	18	6	10	34	65
15-2	WATER	76.5	6	10	42	65	52-1	WATER	80	5	10	43	65
16	WATER	47	6.5	10	34	65	52-2	WATER	80	5	10	43	65
17	WATER	47	6	10	42	65	52-3	WATER	80	5	10	43	65
18	WATER	47	6.5	10	34	65	52-4	WATER	18	5.7	10	40	65
19	WATER	47	6	10	42	65	52-5	WATER	18	5.7	10	40	65
20-1	WATER	40	6.5	10	34	65	52-6	WATER	18	5.7	10	40	65
20-2	WATER	40	6.5	10	34	65	53	WATER	200	5.7	10	43	65
21-1	WATER	40	6	10	42	65	54	WATER	200	5.7	10	43	65
21-2	WATER	40	6	10	42	65	55	WATER	70	5	10	43	65
22-1	WATER	39.6	6.5	10	34	65	56	WATER	270	5	10	43	65
22-2	WATER	39.6	6.5	10	34	65	57	WATER	20	6.5	10	34	65
23	WATER	79.2	6	10	43	65	57-1	WATER	8	6.5	10	34	65
23-1	WATER	39.6	6	10	43	65	57-2	WATER	12	6.5	10	34	65
23-2	WATER	39.6	6	10	43	65	58-1	WATER	8	6	10	62	80
24	WATER	30.5	6.5	10	34	65	58-2	WATER	12	6	10	62	80
24-1	WATER	17.5	6.5	10	34	65	59	WATER	0.3	6	10	37	65
24-2	WATER	13	6.5	10	34	65	60	WATER	0.3	5.7	10	43	65
25	WATER	30.5	6	10	43	65							
25-1	WATER	17.5	6	10	43	65							
25-2	WATER	13	6	10	43	65							
26	WATER	110	6	10	43	65							
27	WATER	200	6	10	43	65							
28	WATER	400	6	10	43	65							
29	WATER	740	6	10	43	65							
30	WATER	80	6.5	10	34	65							
31	WATER	20	6.5	10	34	65							
31-1	WATER	8	6.5	10	34	65							
31-2	WATER	12	6.5	10	34	65							
32	WATER	20	6.5	10	34	65							
32-1	WATER	8	6.5	10	34	65							
32-2	WATER	12	6.5	10	34	65							
33	WATER	20	6.5	10	34	65							
33-1	WATER	8	6.5	10	34	65							
33-2	WATER	12	6.5	10	34	65							
34	WATER	6	6	10	34	65							
34-1	WATER	2	6	10	34	65							
35	WATER	59	6	10	34	65							
36	WATER	270	6	10	34	65							
37-1	WATER	8	6	10	62	80							
37-2	WATER	12	6	10	62	80							
38-1	WATER	8	6	10	62	80							
38-2	WATER	12	6	10	62	80							
39-1	WATER	8	6	10	62	80							
39-2	WATER	12	6	10	62	80							
40	WATER	6	4	10	37	80							

REV.	DATE	DWN	DGND	CHKD	APPROVALS						ISSUED FOR
					CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	
0	2015-06-30	LJCh	CLLi	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	YWHu	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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PLACE	:
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SHEET CONTENT

MASS BALANCE DIAGRAM AUXILIARY COOLING WATER SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT

LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

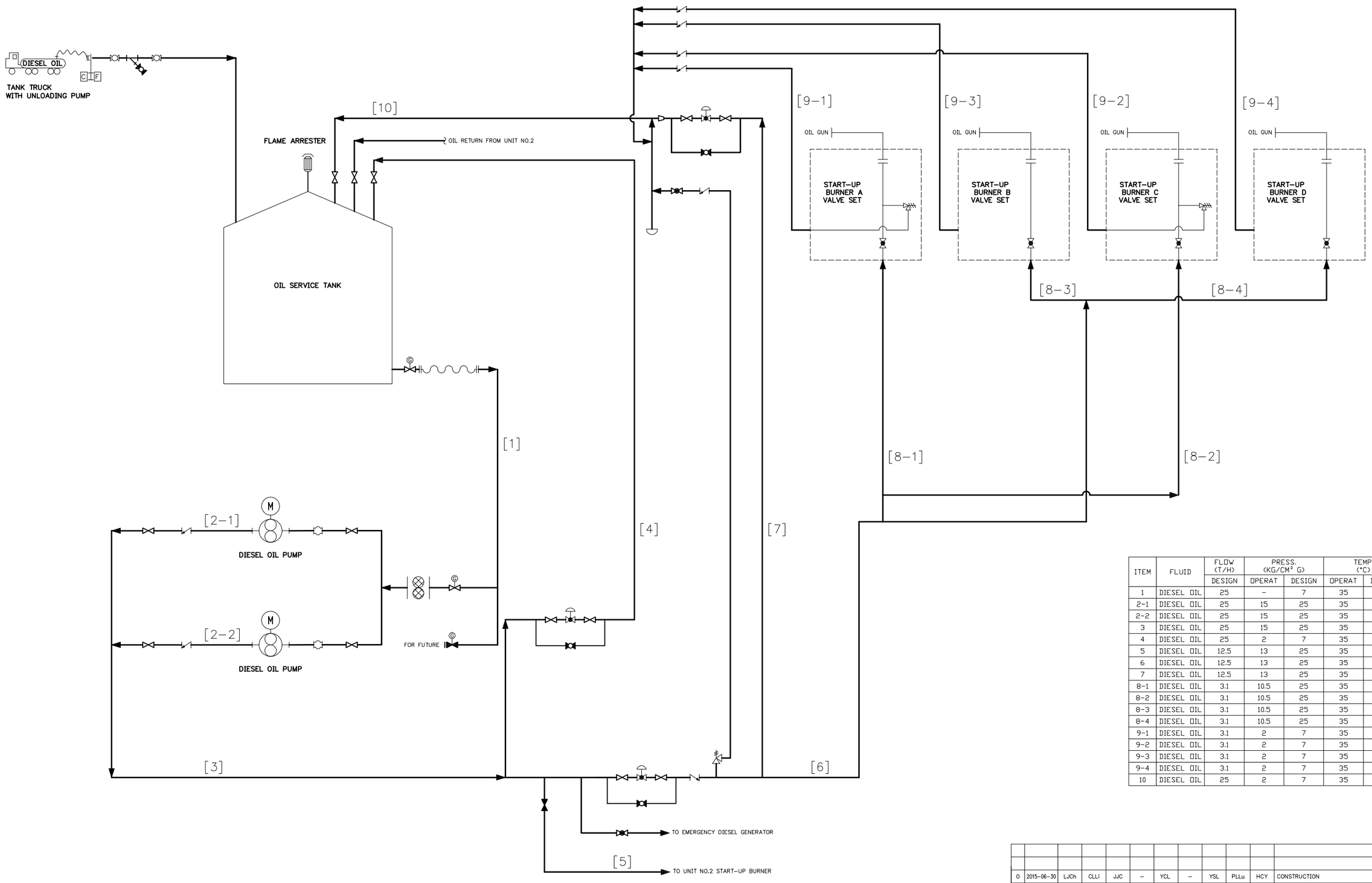
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-021	REV.	SCALE	NONE	SHT.NO.
		0	JOB NO.	1881	1/1

Drawing name: D:\draw\1881\ k1au3-mb-021-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	DIESEL OIL	25	–	7	35	70
2-1	DIESEL OIL	25	15	25	35	70
2-2	DIESEL OIL	25	15	25	35	70
3	DIESEL OIL	25	15	25	35	70
4	DIESEL OIL	25	2	7	35	70
5	DIESEL OIL	12.5	13	25	35	70
6	DIESEL OIL	12.5	13	25	35	70
7	DIESEL OIL	12.5	13	25	35	70
8-1	DIESEL OIL	3.1	10.5	25	35	70
8-2	DIESEL OIL	3.1	10.5	25	35	70
8-3	DIESEL OIL	3.1	10.5	25	35	70
8-4	DIESEL OIL	3.1	10.5	25	35	70
9-1	DIESEL OIL	3.1	2	7	35	70
9-2	DIESEL OIL	3.1	2	7	35	70
9-3	DIESEL OIL	3.1	2	7	35	70
9-4	DIESEL OIL	3.1	2	7	35	70
10	DIESEL OIL	25	2	7	35	70

REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
					APPROVALS						
0	2015-06-30	LJCh	CLLi	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	CLLi	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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REG. NO.	:
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SHEET CONTENT

MASS BALANCE DIAGRAM
DIESEL OIL SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT

LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

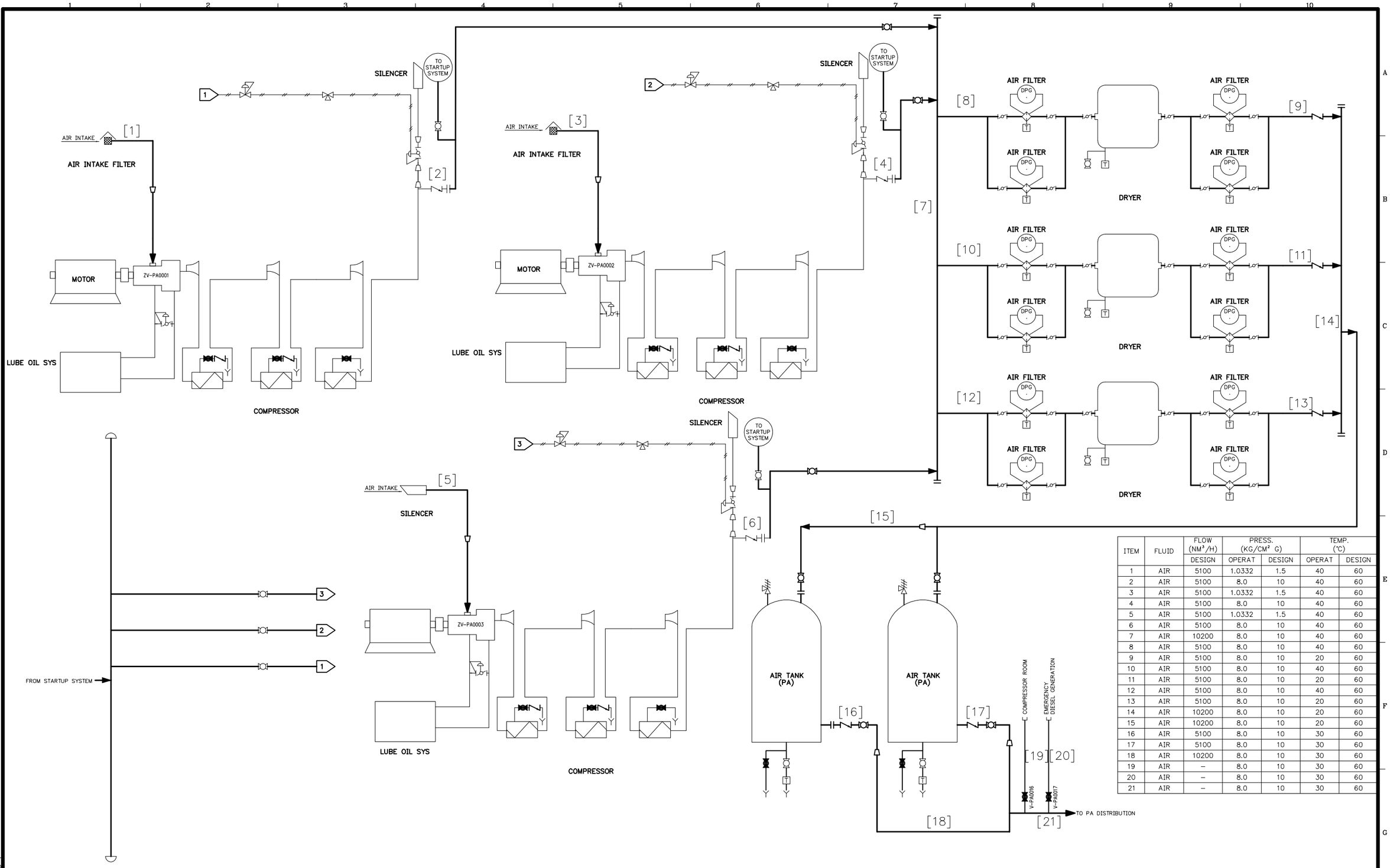
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40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-022	REV.	0
		SCALE	NONE
		JOB NO.	1881
		SHT.NO.	1/1

Drawing name: D:\draw\1881\ k1au3-mb-022-0.dwg Dated: 2015-07-07 4:45 pm By : ljch

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ITEM	FLUID	FLOW (NM^3/H)	PRESS. (KG/CM^2 G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	AIR	5100	1.0332	1.5	40	60
2	AIR	5100	8.0	10	40	60
3	AIR	5100	1.0332	1.5	40	60
4	AIR	5100	8.0	10	40	60
5	AIR	5100	1.0332	1.5	40	60
6	AIR	5100	8.0	10	40	60
7	AIR	10200	8.0	10	40	60
8	AIR	5100	8.0	10	40	60
9	AIR	5100	8.0	10	20	60
10	AIR	5100	8.0	10	40	60
11	AIR	5100	8.0	10	20	60
12	AIR	5100	8.0	10	40	60
13	AIR	5100	8.0	10	20	60
14	AIR	10200	8.0	10	20	60
15	AIR	10200	8.0	10	20	60
16	AIR	5100	8.0	10	30	60
17	AIR	5100	8.0	10	30	60
18	AIR	10200	8.0	10	30	60
19	AIR	–	8.0	10	30	60
20	AIR	–	8.0	10	30	60
21	AIR	–	8.0	10	30	60

REV.	DATE	DWN	DGND	CHKD	APPROVALS CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
0	2015-06-30	LJCh	DJC	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	CSW	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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P.T.R. NO. :
DATE :
PLACE :
T.I.N. :

SHEET CONTENT

MASS BALANCE DIAGRAM COMPRESSED AIR SYSTEM (PLANT AIR)

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT

LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

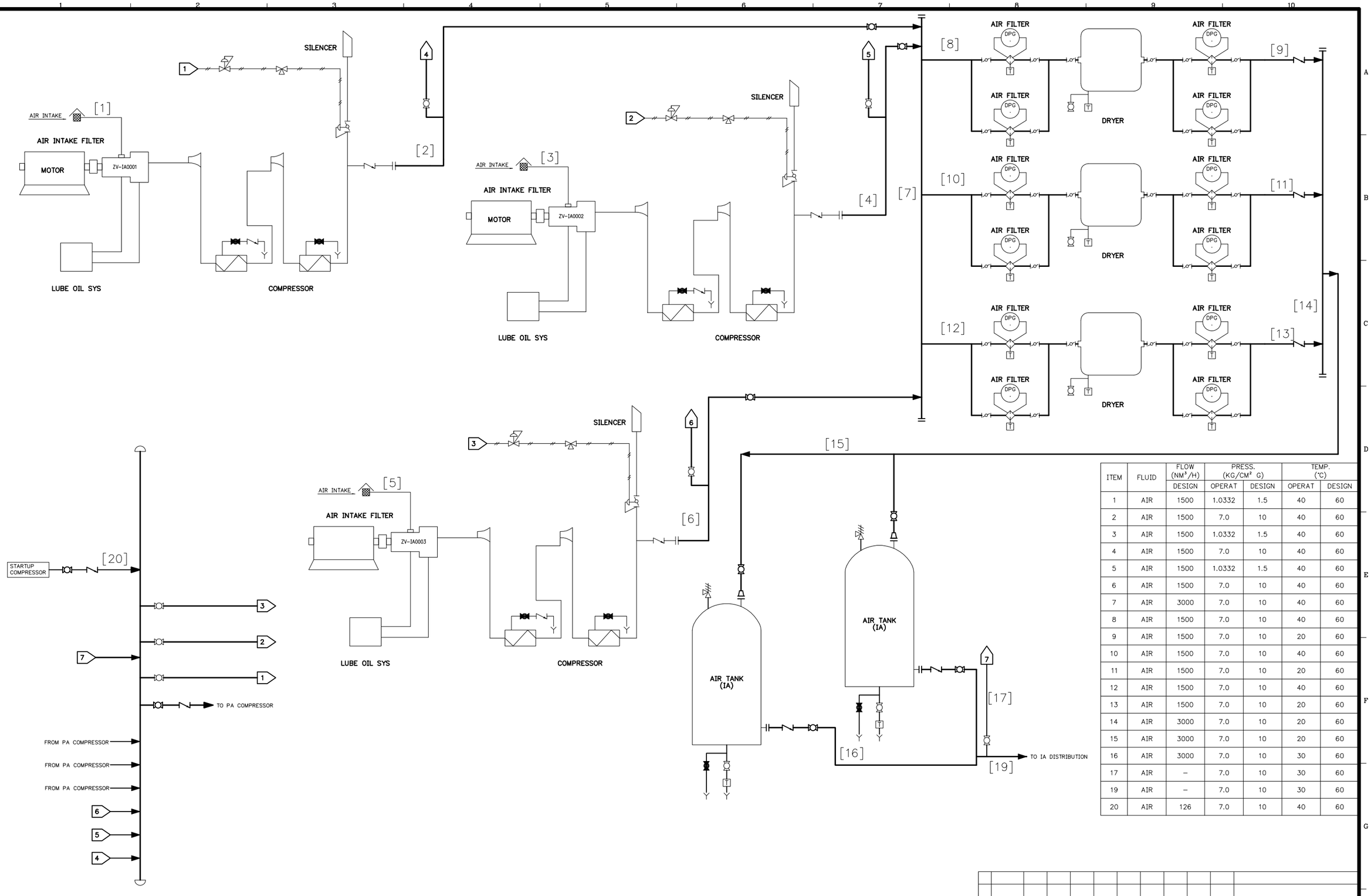
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO. K1AU3-MB-023 | REV. 0 | SCALE NONE | JOB NO. 1881 | SHT.NO. 1/2

Drawing name: D:\draw\1881\ k1au3-mb-023-01-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (NM^3/H)	PRESS. (KG/CM^2 G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	AIR	1500	1.0332	1.5	40	60
2	AIR	1500	7.0	10	40	60
3	AIR	1500	1.0332	1.5	40	60
4	AIR	1500	7.0	10	40	60
5	AIR	1500	1.0332	1.5	40	60
6	AIR	1500	7.0	10	40	60
7	AIR	3000	7.0	10	40	60
8	AIR	1500	7.0	10	40	60
9	AIR	1500	7.0	10	20	60
10	AIR	1500	7.0	10	40	60
11	AIR	1500	7.0	10	20	60
12	AIR	1500	7.0	10	40	60
13	AIR	1500	7.0	10	20	60
14	AIR	3000	7.0	10	20	60
15	AIR	3000	7.0	10	20	60
16	AIR	3000	7.0	10	30	60
17	AIR	–	7.0	10	30	60
19	AIR	–	7.0	10	30	60
20	AIR	126	7.0	10	40	60

REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	
					APPROVALS						ISSUED FOR
0	2015-06-30	LJCh	DJC	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	CSW	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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P.T.R. NO. :
DATE :
PLACE :
T.I.N. :

SHEET CONTENT

MASS BALANCE DIAGRAM COMPRESSED AIR SYSTEM (INSTRUMENT AIR)

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

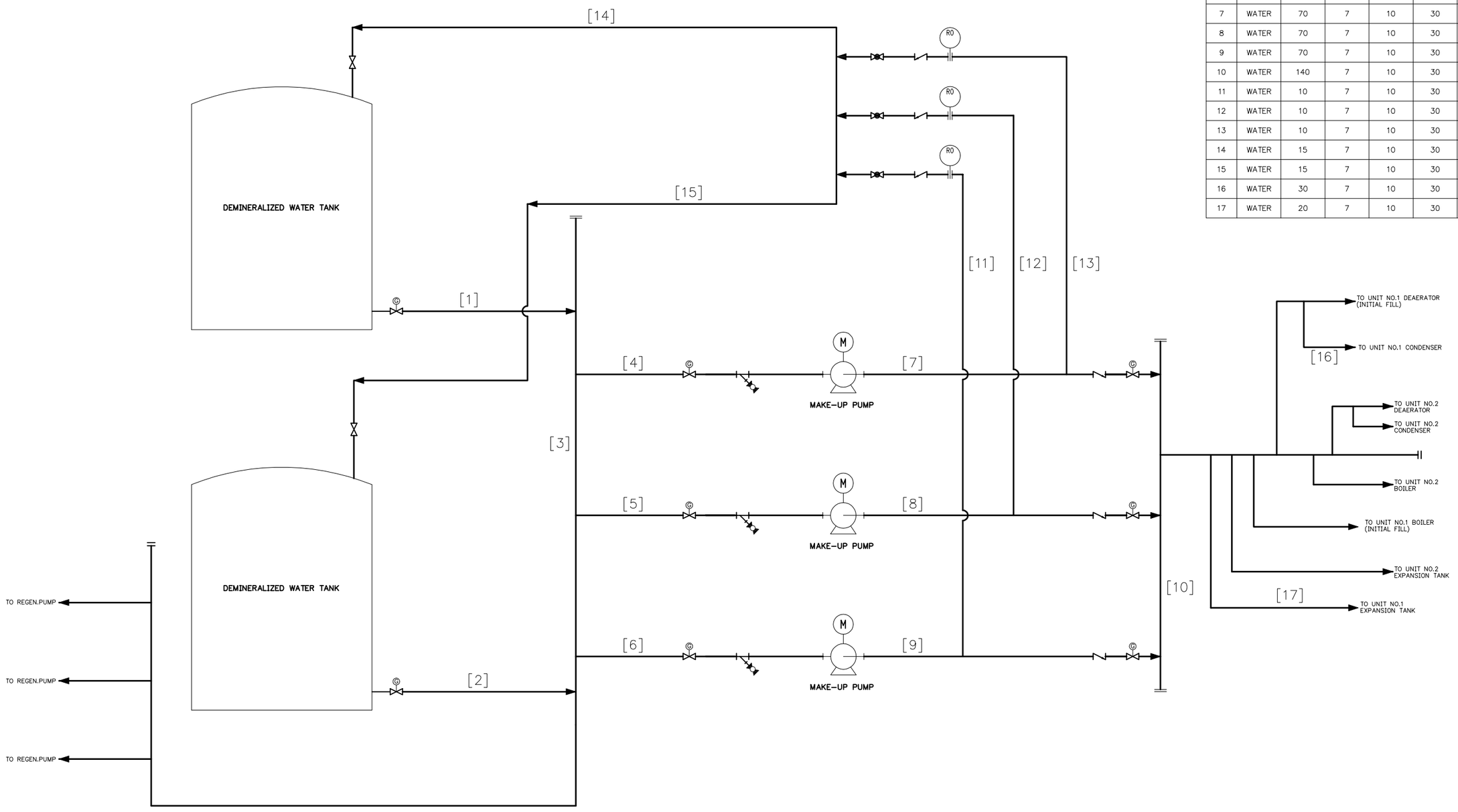
吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO. K1AU3-MB-023 | REV. 0 | SCALE NONE | SHT.NO. 2/2 | JOB NO. 1881

Drawing name: D:\draw\1881\ k1au3-mb-023-02-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	WATER	140	1	1.5	30	65
2	WATER	140	1	1.5	30	65
3	WATER	140	1	1.5	30	65
4	WATER	70	1	1.5	30	65
5	WATER	70	1	1.5	30	65
6	WATER	70	1	1.5	30	65
7	WATER	70	7	10	30	65
8	WATER	70	7	10	30	65
9	WATER	70	7	10	30	65
10	WATER	140	7	10	30	65
11	WATER	10	7	10	30	65
12	WATER	10	7	10	30	65
13	WATER	10	7	10	30	65
14	WATER	15	7	10	30	65
15	WATER	15	7	10	30	65
16	WATER	30	7	10	30	65
17	WATER	20	7	10	30	65



REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
					APPROVALS						
0	2015-06-30	LJCh	WH	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	WH	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

COPY

ENGINEER

REG. NO.	:
P.T.R. NO.	:
DATE	:
PLACE	:
T.I.N.	:

SHEET CONTENT

MASS BALANCE DIAGRAM
MAKE-UP WATER SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

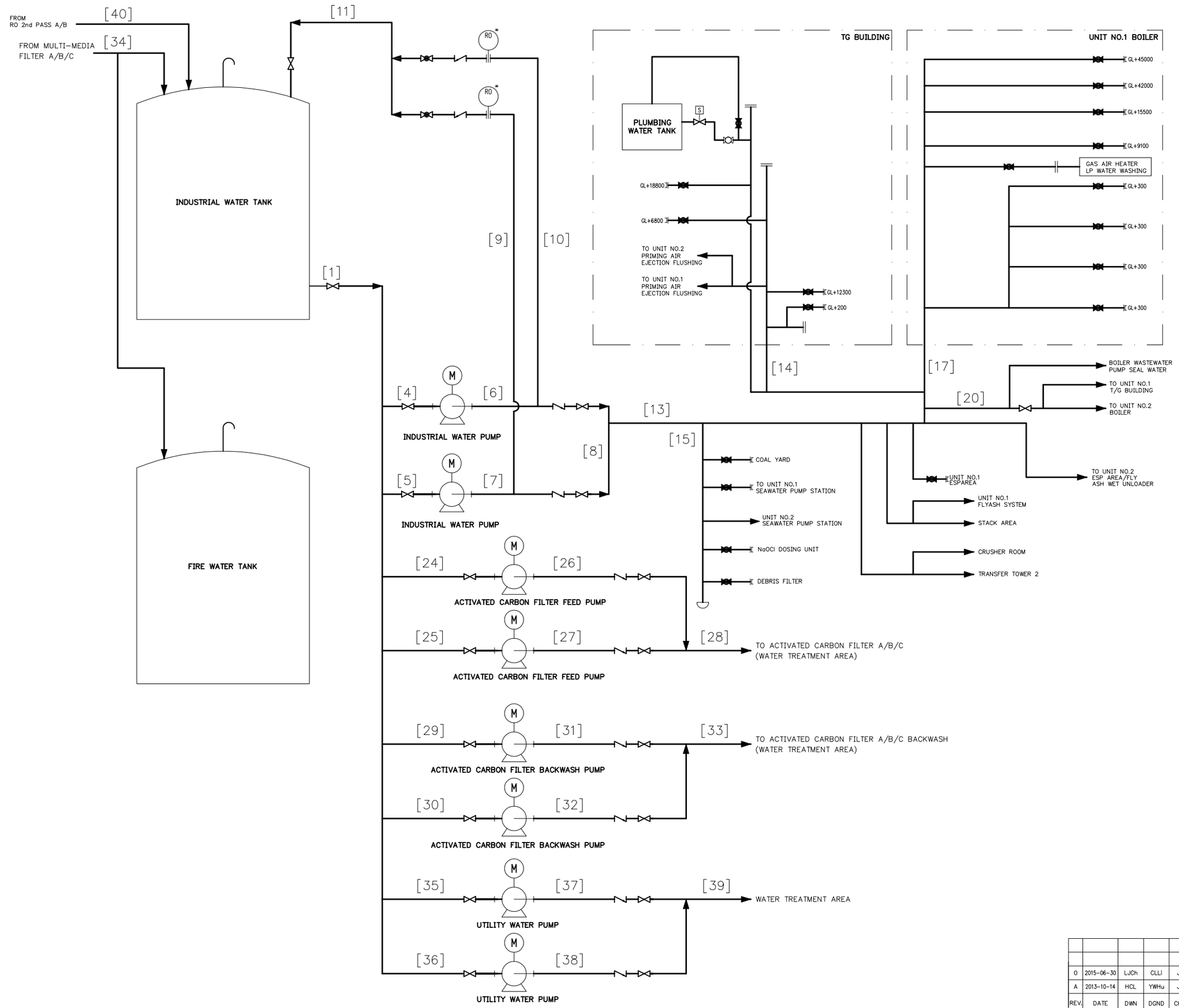
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-024	REV.	0
		SCALE	NONE
		JOB NO.	1881
		SHT.NO.	1/1

Drawing name: D:\draw\1881\ k1au3-mb-024-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	WATER	351	1	1.5	30	65
4	WATER	45	1	1.5	30	65
5	WATER	45	1	1.5	30	65
6	WATER	45	7	10	30	65
7	WATER	45	7	10	30	65
8	WATER	90	7	10	30	65
9	WATER	5	7	10	30	65
10	WATER	5	7	10	30	65
11	WATER	10	7	10	30	65
13	WATER	90	7	10	30	65
14	WATER	10	5	10	30	65
15	WATER	10	5	10	30	65
17	WATER	10	5	10	30	65
20	WATER	10	5	10	30	65
24	WATER	34	1	1.5	30	65
25	WATER	34	1	1.5	30	65
26	WATER	34	5	7.5	30	65
27	WATER	34	5	7.5	30	65
28	WATER	68	5	7.5	30	65
29	WATER	190	1	1.5	30	65
30	WATER	190	1	1.5	30	65
31	WATER	190	3	4.5	30	65
32	WATER	190	3	4.5	30	65
33	WATER	190	3	4.5	30	65
34	WATER	126	1	1.5	30	65
35	WATER	3	1	1.5	30	65
36	WATER	3	1	1.5	30	65
37	WATER	3	4	6	30	65
38	WATER	3	4	6	30	65
39	WATER	3	4	6	30	65

REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
					APPROVALS						
0	2015-06-30	LJCh	CLLi	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	YWHu	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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ENGINEER

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P.T.R. NO. :
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SHEET CONTENT

MASS BALANCE DIAGRAM
INDUSTRIAL WATER SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

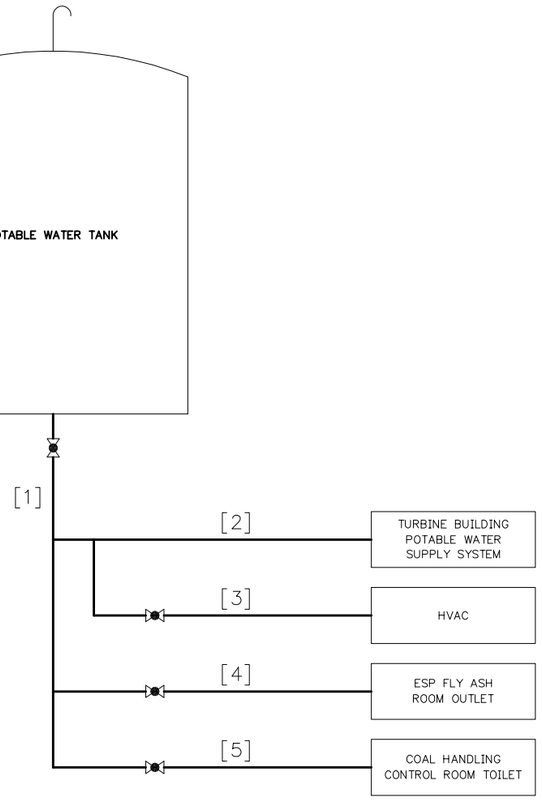
吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO. K1AU3-MB-025 | REV. 0 | SCALE NONE | JOB NO. 1881 | SHT.NO. 1/1

Drawing name: D:\draw\1881\ k1au3-mb-025-0.dwg Dated: 2015-07-07 4:45 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	WATER	10	2	7	30	65
2	WATER	5	2	7	30	65
3	WATER	10	1.5	7	30	65
4	WATER	5	1.5	7	30	65
5	WATER	5	1.5	7	30	65



REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
					APPROVALS						
0	2015-06-30	LJCh	CLLi	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	YWHu	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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REG. NO.	:
P.T.R. NO.	:
DATE	:
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T.I.N.	:

SHEET CONTENT

MASS BALANCE DIAGRAM POTABLE WATER SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT

LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation

40 SAN MIGUEL AVENUE, MANDALUYONG CITY

1550 METRO MANILA, PHILIPPINES

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GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-026	REV.	SCALE	NONE	SHT.NO.
		0	JOB NO.	1881	1/1

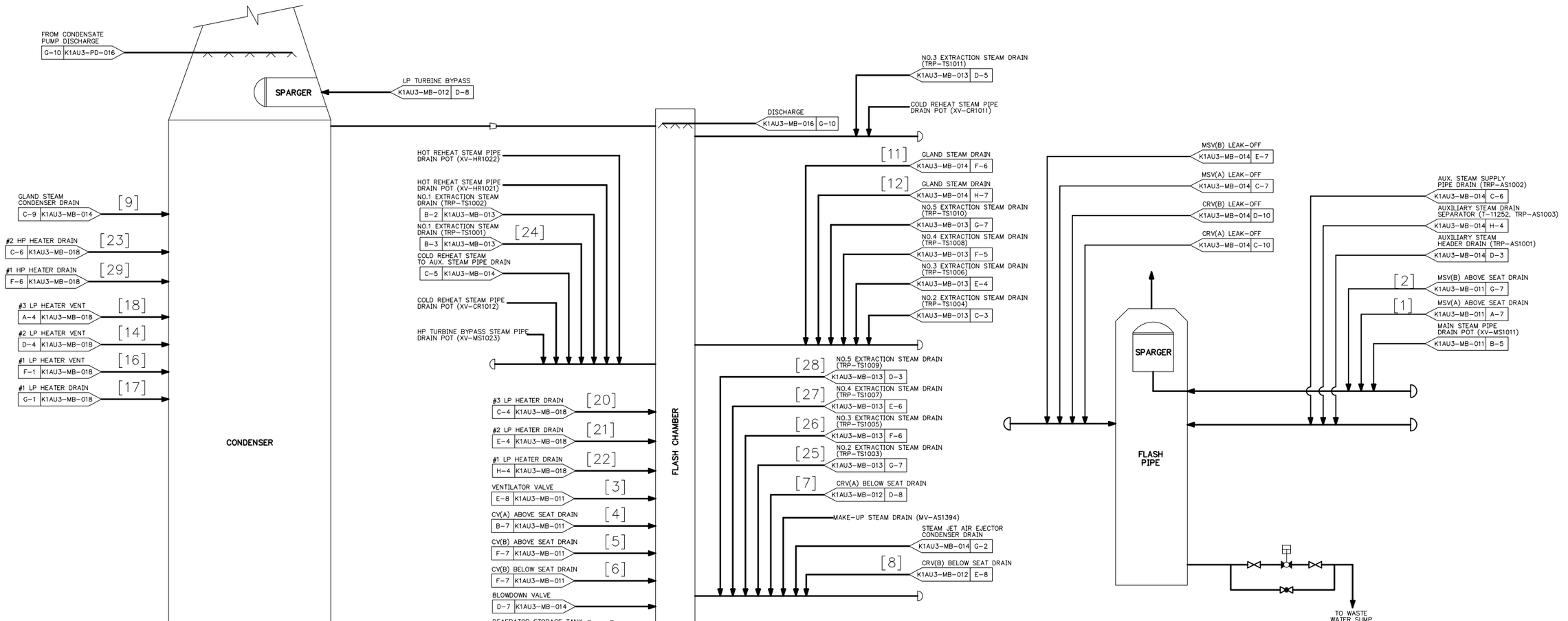
Drawing name: D:\draw\1881\ k1au3-mb-026-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	WATER	–	126	132	538	550
2	WATER	–	126	132	538	550
3	STEAM	–	38	45	368	395
4	WATER	–	126	132	538	550
5	WATER	–	126	132	538	550
6	WATER	–	126	132	538	550
7	WATER	–	38	45	538	550
8	WATER	–	38	45	538	550
9	WATER	–	0.3	2&F.V.	140	400
11	WATER	–	0.3	2&F.V.	140	400
12	WATER	–	0.3	2&F.V.	140	400
13	WATER	–	5.05	8.0	158.4	170
14	AIR& VAPOR	–	−0.215	1.055&F.V.	76	150

ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
16	AIR& VAPOR	–	−0.646	1.055&F.V.	45	150
17	WATER	45.377	−0.646	1.055&F.V.	45	150
18	AIR& VAPOR	–	0.82	1.055&F.V.	96	200
19	WATER	–	0.3	2&F.V.	260	400
20	WATER	15.246	−0.93	1.055&F.V.	96	200
21	WATER	27.516	−0.93	1.055&F.V.	76	150
22	WATER	45.377	−0.93	1.055&F.V.	45	150
23	WATER	45.147	−0.93	1.055&F.V.	201	260
24	WATER	–	38	45	370	395
25	WATER	–	13.87	17.6	417	450
26	WATER	–	5.37	7	310	330
27	WATER	–	0.91	3.5	181	260
28	WATER	–	−0.17	3.5	105	140

ITEM	FLUID	FLOW (T/H)	PRESS. (KG/CM² G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
29	WATER	69.347	−0.93	1.055&F.V.	166	210



REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
					APPROVALS						
0	2015-06-30	LJCh	WH	JJC	–	YCL	–	YSL	PLLu	HCY	CONSTRUCTION
A	2013-10-14	HCL	WH	JJC	–	YCL	–	YSL	PLLu	HCY	INFORMATION

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P.T.R. NO. :
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SHEET CONTENT

MASS BALANCE DIAGRAM MISCELLANEOUS VENT AND DRAIN SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT

LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

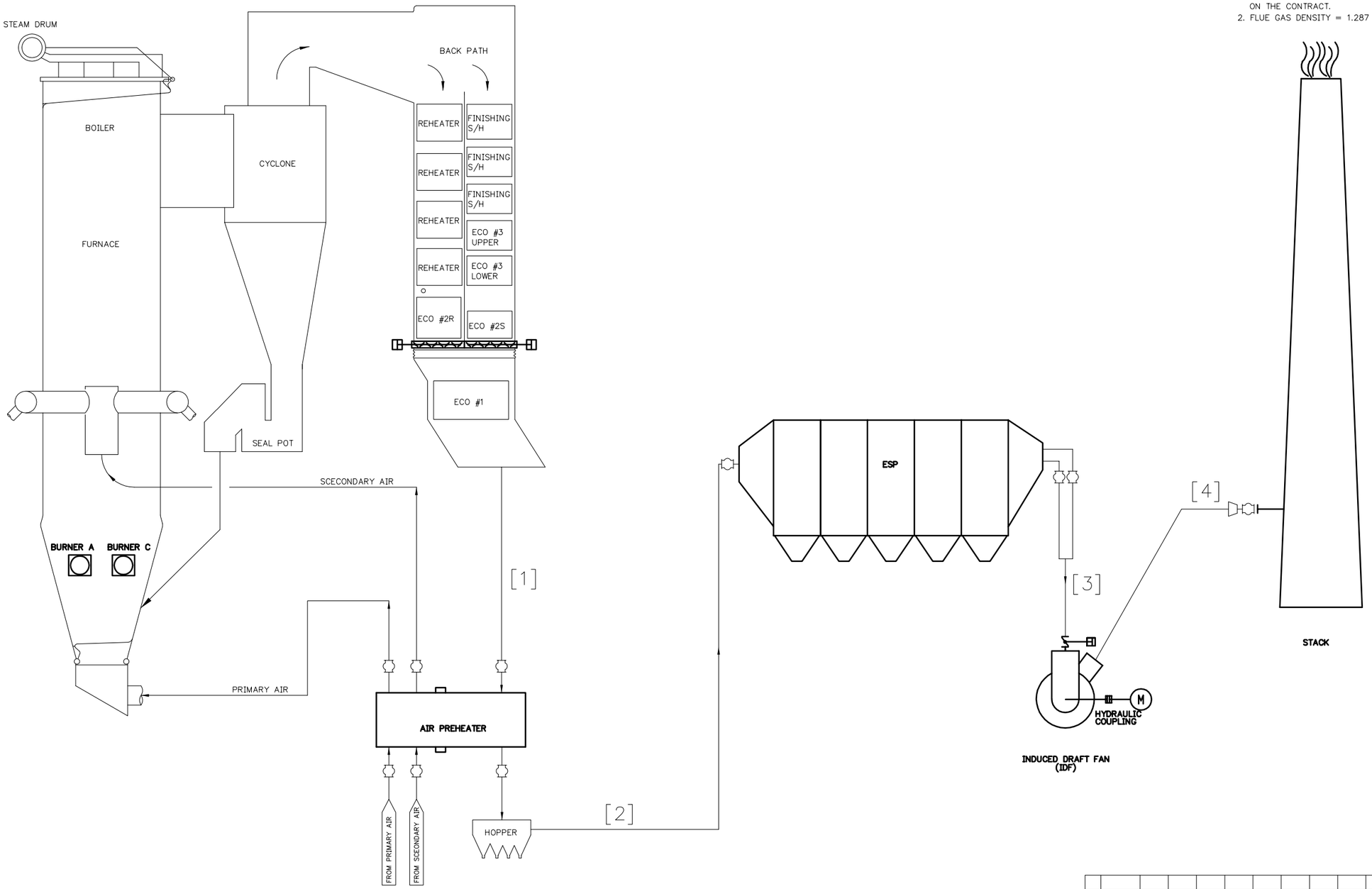
DWG NO.	K1AU3-MB-027	REV.	SCALE	NONE	SHT.NO.
		0	JOB NO.	1881	1/1

Drawing name: D:\draw\1881\ k1au3-mb-027-0.dwg Dated: 2015-07-01 4:43 pm By : ljch

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ITEM	FLUID	FLOW (T/H)	PRESS. (mmH_2O G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	FLUE GAS	679	-420	+890 -890	322	644
2	FLUE GAS	732	-540	+890 -890	139	200
3	FLUE GAS	732	-565	+890 -890	139	200
4	FLUE GAS	732	25	+890 -890	139	200

REMARK:
1. ALL DATA BASED ON 100% BMCR & DESIGN COAL SPECIFIED ON THE CONTRACT.
2. FLUE GAS DENSITY = 1.287 KG/NM^3 .



REV.	DATE	DWN	DGND	CHKD	APPROVALS					ISSUED FOR
					CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	
A	2015-06-30	LJCh	HGL	JJC	–	YCL	–	YSL	PLLu	HCY	REVIEW & COMMENT

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ENGINEER

REG. NO. :
P.T.R. NO. :
DATE :
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SHEET CONTENT

MASS BALANCE DIAGRAM COMBUSION AIR / FLUE GAS SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

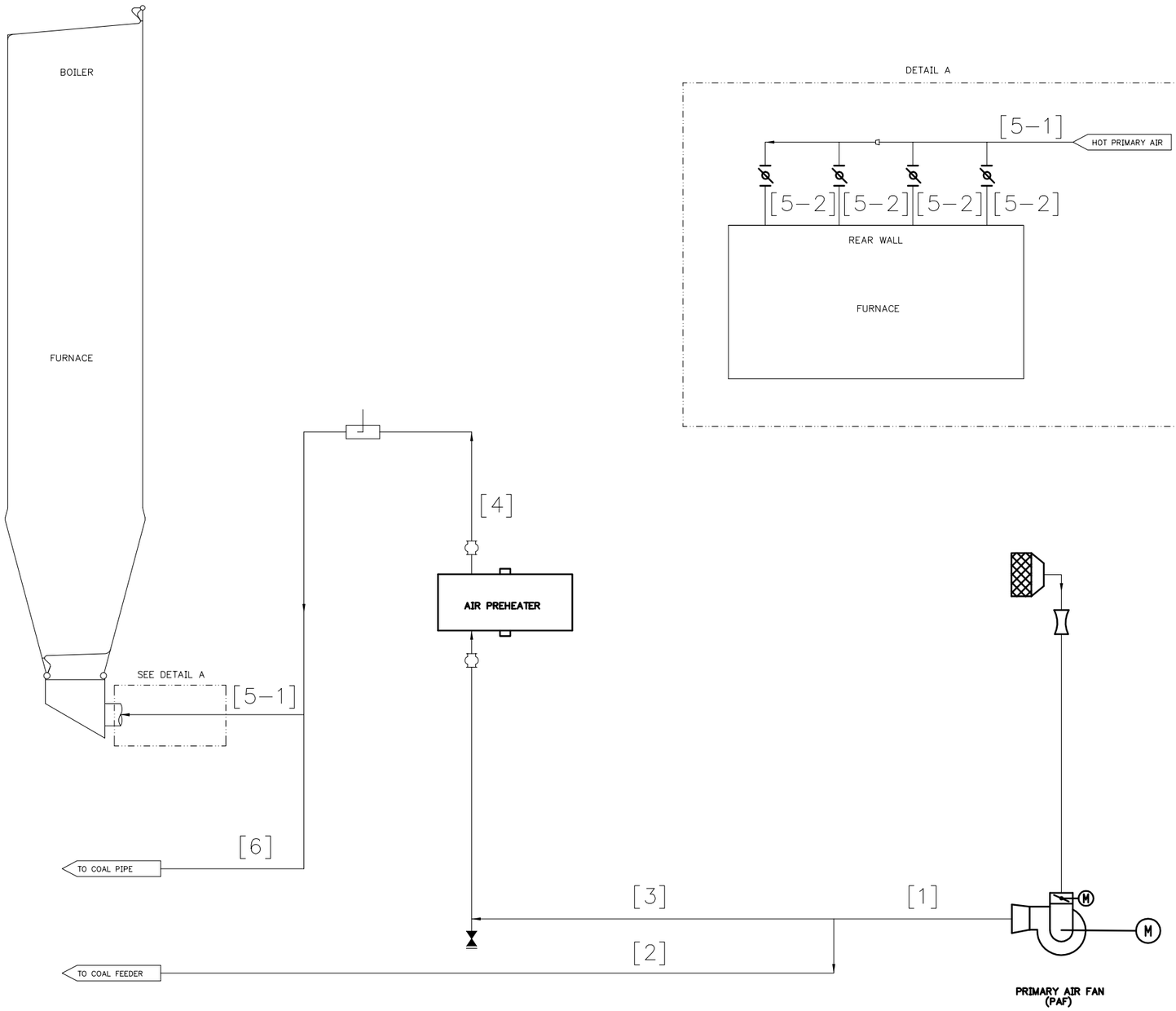
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GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	REV.	SCALE	SHT.NO.
K1AU3-MB-051	A	NONE	1/1
		JOB NO. 1881	

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ITEM	FLUID	FLOW (T/H)	PRESS. (mmH_2O G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	AIR	388	1862	+2500 −340	51	79.4
2	AIR	3	1862	+2500 −340	51	79.4
3	AIR	385	1862	+2500 −340	51	79.4
4	AIR	352	1771	+2500 −340	287	343
5-1	AIR	325	1642	+2500 −340	287	343
5-2	AIR	81.25	1642	+2500 −340	287	343
6	AIR	27	1642	+2500 −340	287	343

REMARK:
1. ALL DATA BASED ON 100% BMCR & DESIGN COAL SPECIFIED ON THE CONTRACT.
2. AIR DENSITY = 1.287 KG/NM³ .



DETAIL A

REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	
					APPROVALS						
A	2015-06-30	LJCh	HGL	JJC	–	YCL	–	YSL	PLLu	HCY	REVIEW & COMMENT

ISSUED FOR

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P.T.R. NO. :
DATE :
PLACE :
T.I.N. :

SHEET CONTENT

MASS BALANCE DIAGRAM
PRIMARY AIR SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

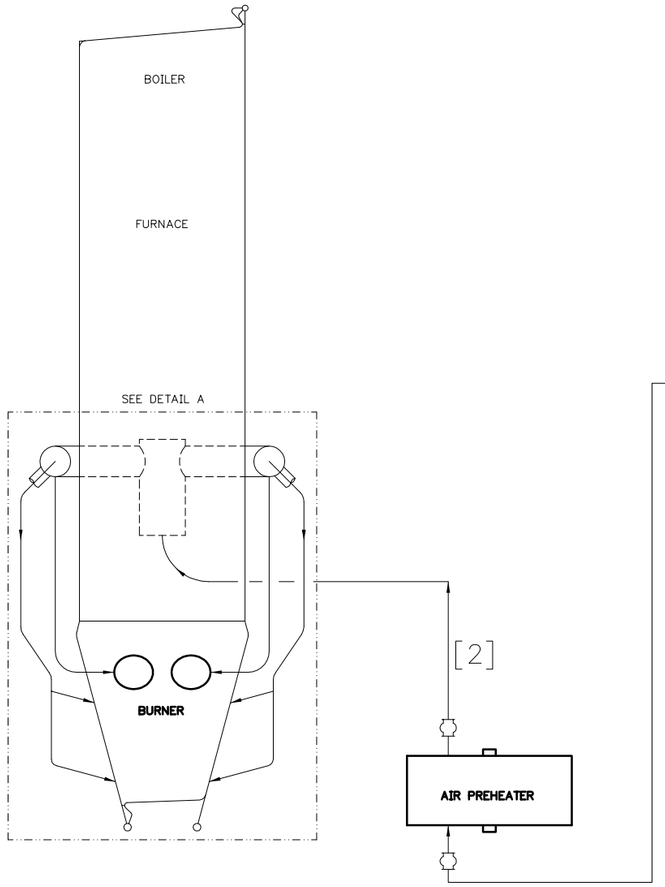
吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

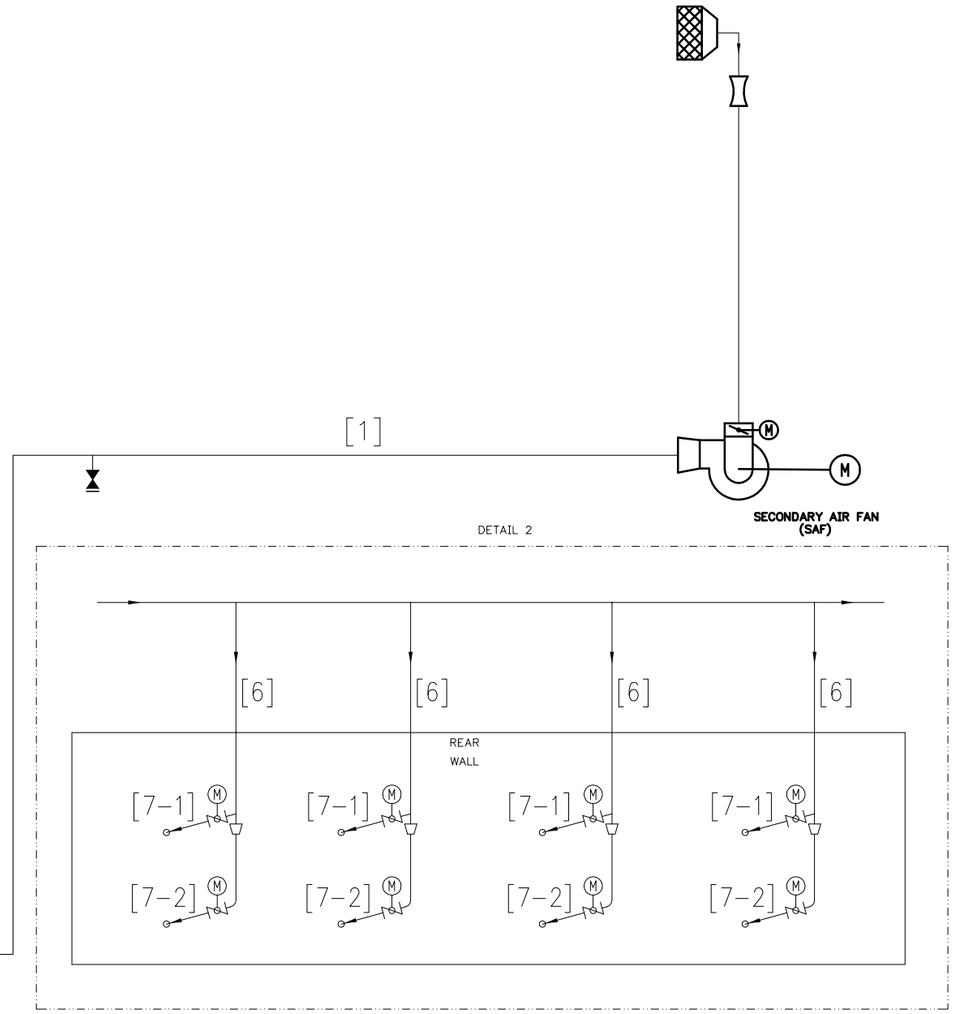
DWG NO.	K1AU3-MB-052	REV.	SCALE	NONE	SHT.NO.
		A	JOB NO.	1881	1/1

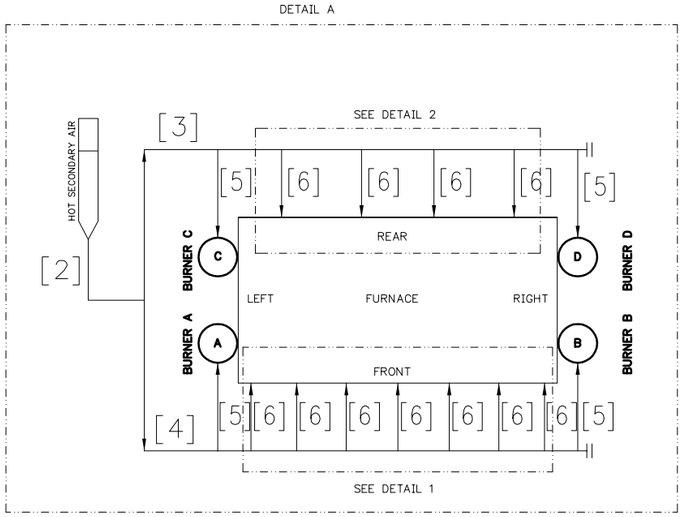
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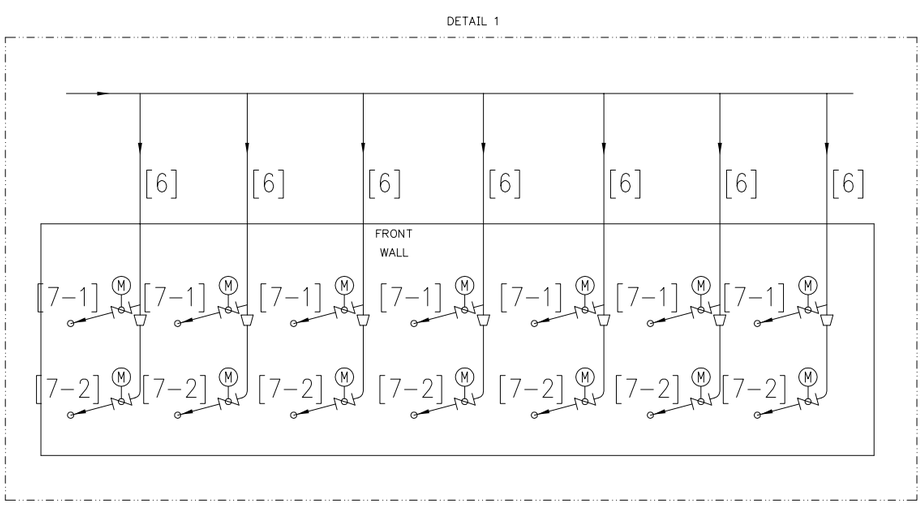
ITEM	FLUID	FLOW (T/H)	PRESS. (mmH_2O G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	AIR	244 (235)	1537	+2000 −340	51	79.4
2	AIR	224 (215)	1442	+2000 −340	302	343
3	AIR	93.2 (106)	1442	+2000 −340	302	343
4	AIR	130.8 (109)	1442	+2000 −340	302	343
5	AIR	7.1 (51)	1266	+2000 −340	302	343
6	AIR	19.6 (1)	1266	+2000 −340	302	343
7-1	AIR	9.8 (0.5)	1266	+2000 −340	302	343
7-2	AIR	9.8 (0.5)	1266	+2000 −340	302	343

REMARK:
1. ALL DATA BASED ON 100% BMCR & DESIGN COAL SPECIFIED ON THE CONTRACT.
2. AIR DENSITY = 1.287 KG/NM^3 .
3. (**) MEANS CONDITION AT FULL OIL FIRING (40% BMCR)









REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	
					APPROVALS						
A	2015-06-30	LJCh	HGL	JJC	–	YCL	–	YSL	PLLu	HCY	REVIEW & COMMENT

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REG. NO.	:
P.T.R. NO.	:
DATE	:
PLACE	:
T.I.N.	:

SHEET CONTENT

MASS BALANCE DIAGRAM
SECONDARY AIR SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

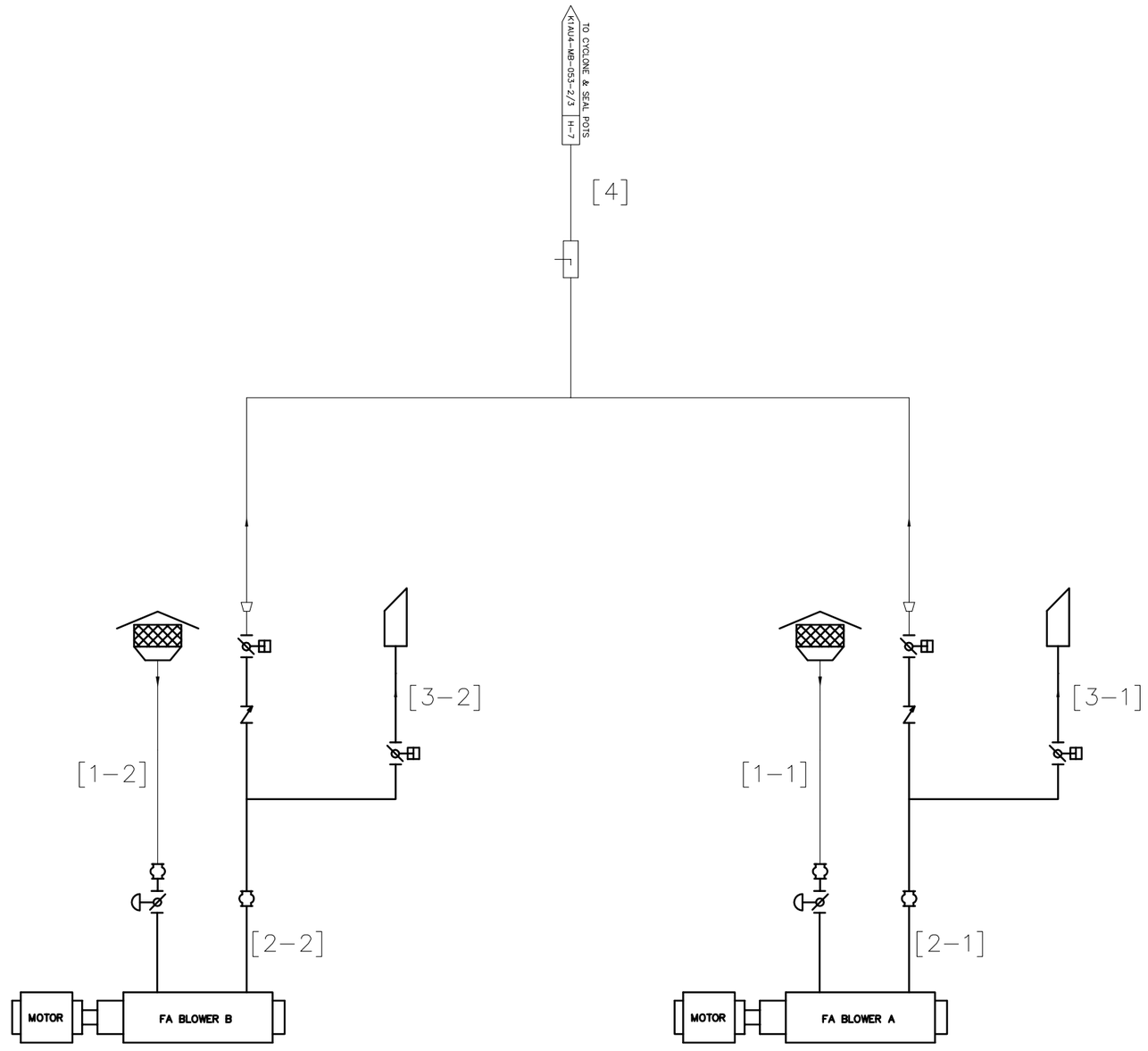
吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-053	REV.	A
		SCALE	NONE
		JOB NO.	1881
		SHT.NO.	1/1

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ITEM	FLUID	FLOW (T/H)	PRESS. (mmH_2O G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1-1	AIR	22.8	--	--	35	65
1-2	AIR	0	--	--	35	65
2-1	AIR	22.8	5230	+8700 -330	91	149
2-2	AIR	0	5230	+8700 -330	91	149
3-1	AIR	0	0	+8700 -330	91	149
3-2	AIR	0	0	+8700 -330	91	149
4	AIR	22.8	5230	+8700 -330	91	149

REMARK:
1. ALL DATA BASED ON 100% BMCR & DESIGN COAL SPECIFIED ON THE CONTRACT.
2. AIR DENSITY = 1.287 KG/NM^3 .



FLUIDIZING AIR BLOWER B
(F.A. BLOWER B)

FLUIDIZING AIR BLOWER A
(F.A. BLOWER A)

REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	
					APPROVALS						
A	2015-06-30	LJCh	HGL	JJC	–	YCL	–	YSL	PLLu	HCY	REVIEW & COMMENT

COPY

ENGINEER

REG. NO. :
P.T.R. NO. :
DATE :
PLACE :
T.I.N. :

SHEET CONTENT
MASS BALANCE DIAGRAM
FLUIDIZING AIR SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

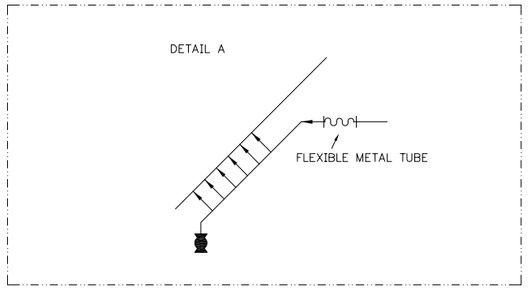
PROJECT TITLE / LOCATION
SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

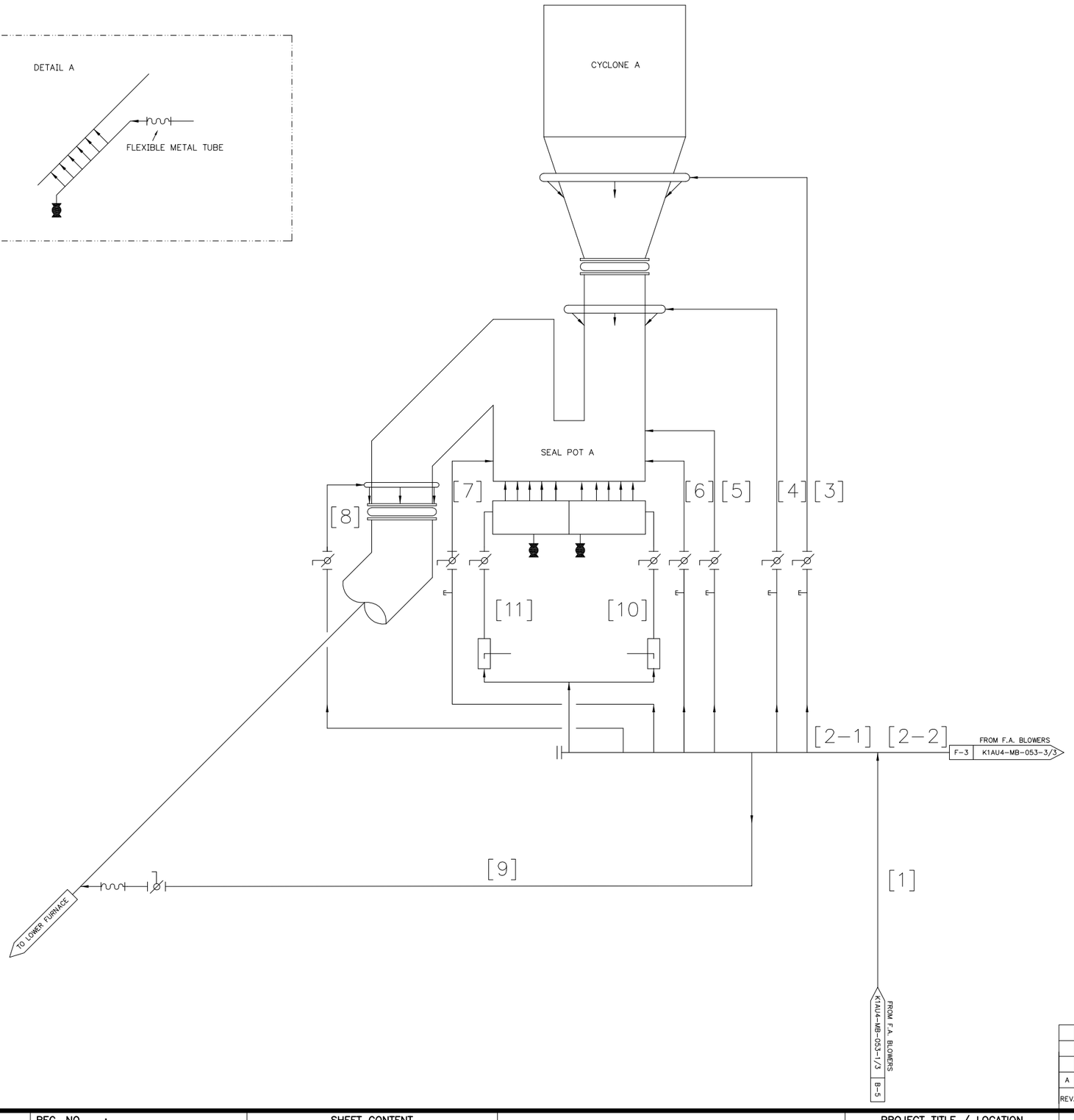
OWNER / ADDRESS
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO. K1AU3-MB-054 | REV. A | SCALE NONE | JOB NO. 1881 | SHT.NO. 1/3

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ITEM	FLUID	FLOW (T/H)	PRESS. (mmH_2O G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	AIR	22.8	5230	+8700 −330	91	149
2-1	AIR	11.4	5230	+8700 −330	91	149
2-2	AIR	11.4	5230	+8700 −330	91	149
3	AIR	1	5230	+8700 −330	91	149
4	AIR	1	5230	+8700 −330	91	149
5	AIR	0.25	5230	+8700 −330	91	149
6	AIR	0.25	5230	+8700 −330	91	149
7	AIR	0.25	5230	+8700 −330	91	149
8	AIR	1	5230	+8700 −330	91	149
9	AIR	0.7	5230	+8700 −330	91	149
10	AIR	3.5	5230	+8700 −330	91	149
11	AIR	3.5	5230	+8700 −330	91	149

REMARK:
1. ALL DATA BASED ON 100% BMCR & DESIGN COAL SPECIFIED ON THE CONTRACT.
2. AIR DENSITY = 1.287 KG/NM^3 .

REV.	DATE	DWN	DGND	CHKD	CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	APPROVALS / ISSUED FOR
A	2015-06-30	LJCh	HGL	JJC	–	YCL	–	YSL	PLLu	HCY	REVIEW & COMMENT

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ENGINEER

REG. NO. :
P.T.R. NO. :
DATE :
PLACE :
T.I.N. :

SHEET CONTENT

MASS BALANCE DIAGRAM
FLUIDIZING AIR SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

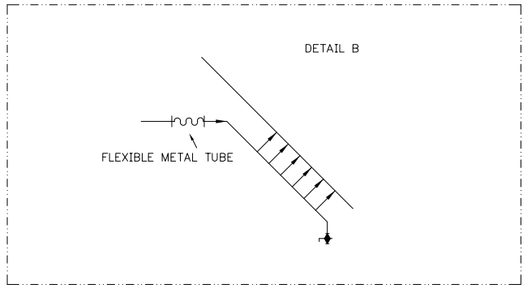
OWNER / ADDRESS

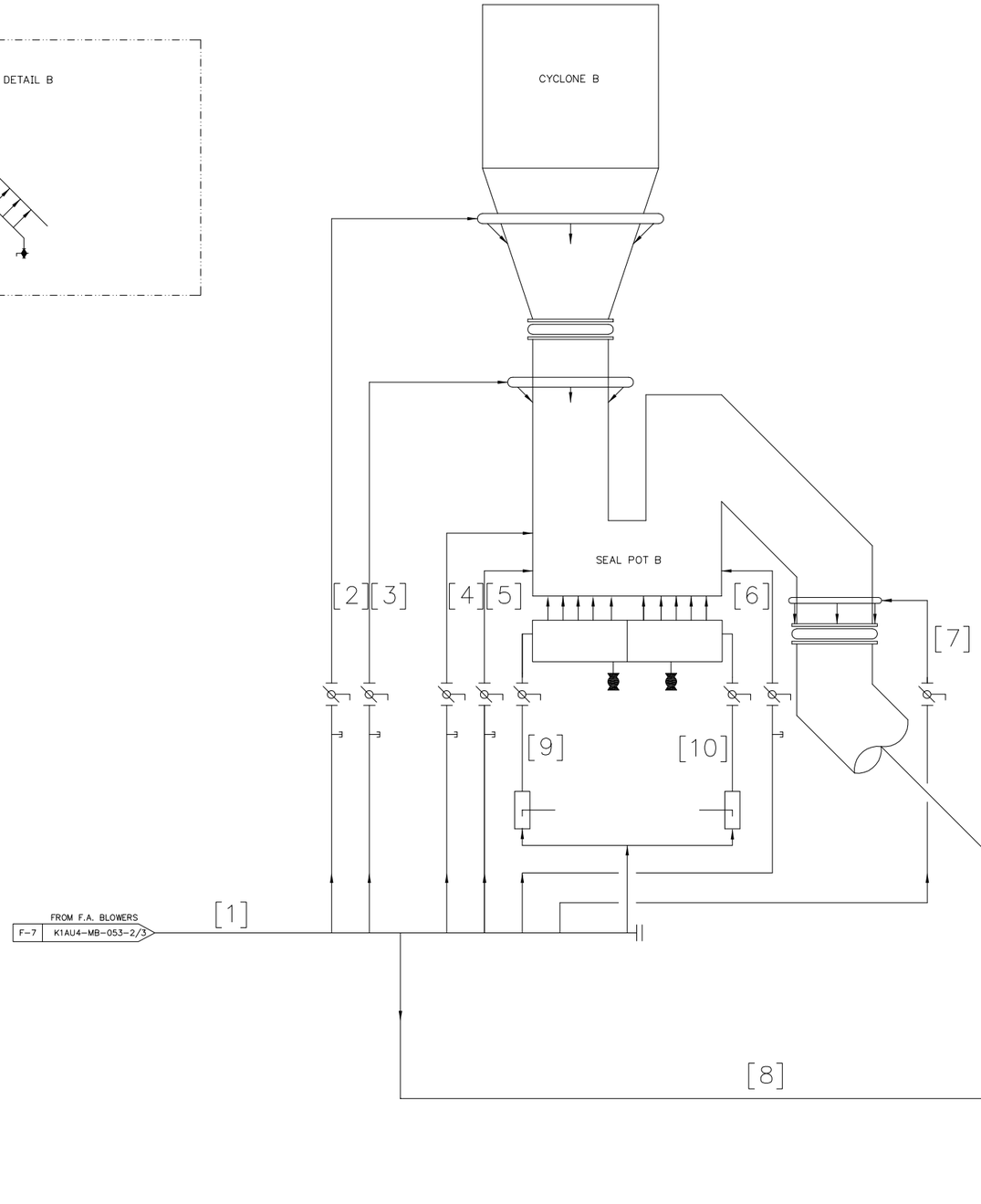
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO. K1AU3-MB-054 | REV. A | SCALE NONE | JOB NO. 1881 | SHT.NO. 2/3

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ITEM	FLUID	FLOW (T/H)	PRESS. (mmH_2O G)		TEMP. (°C)	
		DESIGN	OPERAT	DESIGN	OPERAT	DESIGN
1	AIR	11.4	5230	+8700 −330	91	149
2	AIR	1	5230	+8700 −330	91	149
3	AIR	1	5230	+8700 −330	91	149
4	AIR	0.25	5230	+8700 −330	91	149
5	AIR	0.25	5230	+8700 −330	91	149
6	AIR	0.25	5230	+8700 −330	91	149
7	AIR	1	5230	+8700 −330	91	149
8	AIR	0.7	5230	+8700 −330	91	149
9	AIR	1	5230	+8700 −330	91	149
10	AIR	3.5	5230	+8700 −330	91	149
11	AIR	3.5	5230	+8700 −330	91	149

REMARK:
1. ALL DATA BASED ON 100% BMCR & DESIGN COAL SPECIFIED ON THE CONTRACT.
2. AIR DENSITY = 1.287 KG/NM^3 .

REV.	DATE	DWN	DGND	CHKD	APPROVALS CIVIL	MECH	ELEC	I&C	ENVIR	P.M.	ISSUED FOR
A	2015-06-30	LJCh	HGL	JJC	–	YCL	–	YSL	PLLu	HCY	REVIEW & COMMENT

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ENGINEER

REG. NO. :
P.T.R. NO. :
DATE :
PLACE :
T.I.N. :

SHEET CONTENT

MASS BALANCE DIAGRAM
FLUIDIZING AIR SYSTEM

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

吉興工程顧問股份有限公司
GIBSIN Engineers, Ltd.
TAIPEI, TAIWAN, R.O.C.

DWG NO.	K1AU3-MB-054	REV.	SCALE	NONE	SHT.NO.
		A	JOB NO.	1881	3/3